Central Asia Barometer Data Analysis, continued

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Central Asia Barometer Data Analysis 2 (continuation)

data wrangling, exploratory data analysis, and data visualization of Central Asia Barometer Data

first developed for presentation at American University of Central Asia, Spring 2026

1 Overview

This workshop illustrates working with data, continuing from Central Asia Barometer 1, which introduced working with a single wave and country of Central Asia Barometer data (Kygryzstan in 2022). The second part illustrates working with multiple waves and countries of data, illustrating more complex data merging and cleaning, additional graphics options, and suggests starting points for statistical analysis. Coding is in R, but a Python version of the material (machine-translated) is available for those who wish to replicate the steps there. Both R and Python are excellent tools for data analysis, so to some extent your choice is a matter of preference and familiarity. The R language emerged from statistics and added computing functionality, whereas Python is a general purpose computing language to which heavy doses of statistical capabilities have been added.

2 Installation

This workshop does *not* cover the installation of R and RStudio, but assumes that you have a working copy available.

R is open source software for statistical analysis, available at R-project.org. It is recommended to install R first. R commands can be executed by many editors and IDEs, directly in the terminal, or via script files. However, one of the most powerful and common methods, which we will use here, is to use the RStudio IDE.

RStudio, available from Posit as an open source download, is a tool for creating and running R code (and Python and a few other languages), with many helpful features and templates for working with the other formats and tools in the R ecosystem. RStudio will make use of your

installed version of R, running as a layer on top. Posit is also developing Positron as a newer IDE with more flexibility to work with R, Python, and other possible languages. This workshop does not use Positron, although the code should also work there.

Both RStudio and Positron are introducing AI suggestion tools built into their interfaces that claim to work with and understand the specific code of your project. These should be preferred to using external services like ChatGPT.

For immediate use without installation, try the following:

- The Posit Cloud is also a free service that allows you to use RStudio (with some limitations on time and memory) with the creation of an account.

Please see the Getting Started with R video for more details (or other web tutorials and guides).

Python is also open-source, and can be used with various editors and tools, which are not detailed here.

2.1 R help

The R help system is easy to access. Use a question mark before the name of a specific function to pull up its manual page (*?sample* or *?rnorm*). Use two question marks to search The help tab in the lower right quadrant of RStudio also allows full access to the built in help, including manuals and searching.

Searching the internet for R help can be useful, since there are a wealth of tutorials, blog posts, and other material on general methods and specific R packages. It is helpful to add something like "R statistics" or "R software" or "R package" to your search to distinguish from all other uses of the letter R!

3 Setup and preparing data

3.1 R packages required

We will use the *pak* package for installation as a more complete approach to package management. Replace the pkg commands with *install.packages()* versions if you prefer.

This session relies on the *tidyverse* suite of packages for data manipulation as a preference, although the same tasks could be accomplished in base R using other functions. Install *pak* and *tidyverse* if you don't already have them on your system. We will also need *reticulate* to run Python code chunks.

For graphics, *ggplot2* will do most of the work, but *lattice* and several specialized helper packages will also be used.

```
install.packages("pak", dependencies=TRUE)
library(pak)
pkg_install("tidyverse")
pkg_install("readxl")
pkg_install("reticulate")
```

```r
pkg_install("ggplot2") # including in tidyverse, but just making this
 ↪  explicit
pkg_install("lattice")

# helper packages

pkg_install("ggthemes")
pkg_install("cowplot")
pkg_install("gridExtra")
pkg_install("RColorBrewer")
pkg_install("png")
pkg_install("paletteer")
pkg_install("ggiraph")

# pkg_install("ggridges")
# pkg_install("ggvis")
# pkg_install("gganimate")
# pkg_install("gapminder")
# pkg_install("hexbin")
# pkg_install("gifski")

devtools::session_info()
```

Note that the *update.packages()* command can be run periodically to check for newer versions of any packages installed.

Load the packages up front so that we can work with them.

```r
library(readxl)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```r
library(lattice)
library(ggthemes)
library(cowplot)
```

```
Attaching package: 'cowplot'
```

```
The following object is masked from 'package:ggthemes':

    theme_map

The following object is masked from 'package:lubridate':

    stamp
```

```r
library(gridExtra)
```

```
Attaching package: 'gridExtra'

The following object is masked from 'package:dplyr':

    combine
```

```r
library(RColorBrewer)
library(png)
library(paletteer)
library(ggiraph)
```

3.2 Versions

The links to the right provide alternate modes of accessing this content. You can choose Light or Dark mode HTML, a downloadable PDF, or a Word Docx format file all representing the same content. Code files with R (.R) or Python (.py) are also available, if you'd like to isolate the code from the description.

3.3 Kyrgyz and Russian language

Also, machine-translated versions of this content in Russian and Kyrgyz are linked on the right.

3.4 Python support

Now let's load *reticulate* (for Python support only). We must set the correct location of python corresponding to our system.

A Python file is provided for those who want to try out the same data steps in Python. Note that the Python file was machine-translated from R, and probably needs some work!

```r
# Sys.setenv(RETICULATE_PYTHON = "/usr/bin/python3")
library(reticulate)
use_python("/usr/bin/python3")
```

4 Central Asia Barometer

As discussed in the previous workshop, the Central Asia Barometer project conducts polls and surveys across the countries of Central Asia (Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan, and Uzbekistan) and releases the results on a regular basis. Except for the latest wave of data, the CA Barometer data is freely downloadable from their website after providing a contact e-mail. These surveys examine contemporary political and social attitudes, and allow for longitudinal comparisons over time and cross-country comparisons.

In this workshop, we will combine nine files, the survey results for Kazakhstan, Kyrgyzstan, and Uzbekistan for Waves 4, 8, and 12, representing the Fall of 2018, 2020, and 2022 respectively. While more manageable than attempting to merge all countries over even more waves, this gives us a sense of the greater complexity involved with comparisons across countries and over time.

4.1 Understanding the data

Before worrying about coding or statistical analysis, the first step in working with real-world data is to develop an understanding of its structure and meaning. We want to know what data has been collected, how it has been organized, and think about what kinds of questions it can answer. Only then can we know how to properly proceed with analyzing it.

For CA Barometer, this means looking at many files. As previously, you will need to fill out the form on the CA Barometer site before the links will be accessible. The complete list is below:

2018 - Wave 4

Methods Report

Questionnaires: ENG, Kazakh, Kyrgyz, Uzbek

Data: Kazakhstan, Kyrgyzstan, Uzbekistan

2020 - Wave 8

Methods Report

Questionnaires: ENG, RUS

Data: Kazakhstan, Kyrgyzstan, Uzbekistan

2022 - Wave 12

Methods Report

Questionnaire: ENG

Data: Kazakhstan, Kyrgyzstan, Uzbekistan

5 Download and import data

We download data from Central Asia Barometer using the links above. Note links are additionally provided for Tajikistan and Turkmenistan, but we will not be using them in this workshop.

```
download.file("https://ca-barometer.org/assets/files/projects/cab-survey-wave-12-excel-kyrgyzs
↪  "CAB_2022_KG.csv")
data_22_KG <- read_excel("CAB_2022_KG.csv")

download.file("https://ca-barometer.org/assets/files/projects/cab-survey-wave-12-excel-kazakhs
↪  "CAB_2022_KZ.csv")
data_22_KZ <- read_excel("CAB_2022_KZ.csv")

download.file("https://ca-barometer.org/assets/files/projects/cab-survey-wave-12-excel-uzbekis
↪  "CAB_2022_UZ.csv")
data_22_UZ <- read_excel("CAB_2022_UZ.csv")

download.file("https://ca-barometer.org/assets/files/projects/cab-survey-wave-12-excel-turkmen
↪  "CAB_2022_TM.csv")
data_22_TM <- read_excel("CAB_2022_TM.csv")

download.file("https://ca-barometer.org/assets/files/projects/cab-survey-wave-12-excel-tajikis
↪  "CAB_2022_TJ.csv")
data_22_TJ <- read_excel("CAB_2022_TJ.csv")

# download 2020 data

download.file("https://ca-barometer.org/assets/files/projects/central-asia-barometer-survey-wa
↪  "CAB_2020_KG.csv")
data_20_KG <- read_excel("CAB_2020_KG.csv")

download.file("https://ca-barometer.org/assets/files/projects/central-asia-barometer-survey-wa
↪  "CAB_2020_KZ.csv")
data_20_KZ <- read_excel("CAB_2020_KZ.csv")

download.file("https://ca-barometer.org/assets/files/projects/central-asia-barometer-survey-wa
↪  "CAB_2020_UZ.csv")
data_20_UZ <- read_excel("CAB_2020_UZ.csv")

# no 2020 Turkmenistan data
#
↪  download.file("https://ca-barometer.org/assets/files/projects/central-asia-barometer-surve
↪  "CAB_2020_TM.csv")
# data_20_TM <- read_excel("CAB_2020_TM.csv")

download.file("https://ca-barometer.org/assets/files/projects/central-asia-barometer-survey-wa
↪  "CAB_2020_TJ.csv")
data_20_TJ <- read_excel("CAB_2020_TJ.csv")
```

```
# download 2018 data

download.file("https://ca-barometer.org/assets/files/projects/central-asia-barometer-survey-wa
↪  "CAB_2018_KG.csv")
data_18_KG <- read_excel("CAB_2018_KG.csv")

download.file("https://ca-barometer.org/assets/files/projects/central-asia-barometer-survey-wa
↪  "CAB_2018_KZ.csv")
data_18_KZ <- read_excel("CAB_2018_KZ.csv")

download.file("https://ca-barometer.org/assets/files/projects/central-asia-barometer-survey-wa
↪  "CAB_2018_UZ.csv")
data_18_UZ <- read_excel("CAB_2018_UZ.csv")

download.file("https://ca-barometer.org/assets/files/projects/central-asia-barometer-survey-wa
↪  "CAB_2018_TM.csv")
data_18_TM <- read_excel("CAB_2018_TM.csv")

download.file("https://ca-barometer.org/assets/files/projects/central-asia-barometer-survey-wa
↪  "CAB_2018_TJ.csv")
data_18_TJ <- read_excel("CAB_2018_TJ.csv")
```

6 Merge data

Now we must merge the data. Along the way, we discover some inconsistencies that must be cleaned up in order to merge into one coherent file. For example, Country is sometimes abbreviated, sometimes represented as variable MM4.

```
data_22_KG <- mutate(data_22_KG, Country = 'Kyrgyzstan')
data_20_KG <- mutate(data_20_KG, Country = 'Kyrgyzstan')

# data_18_KG <- mutate(data_18_KG, Country = 'KG')

data_22_KZ <- mutate(data_22_KZ, Country = 'Kazakhstan')
data_20_KZ <- mutate(data_20_KZ, Country = 'Kazakhstan')

# data_18_KZ <- mutate(data_18_KZ, Country = 'KZ')

data_22_UZ <- mutate(data_22_UZ, Country = 'Uzbekistan')
data_20_UZ <- mutate(data_20_UZ, Country = 'Uzbekistan')

# data_18_UZ <- mutate(data_18_UZ, Country = 'UZ')

data_22_TM <- mutate(data_22_TM, Country = 'Turkmenistan')

# data_20_TM <- mutate(data_20_TM, Country = 'TM')
# data_18_TM <- mutate(data_18_TM, Country = 'TM')
```

```
data_22_TJ <- mutate(data_22_TJ, Country = 'Tajikistan')
data_20_TJ <- mutate(data_20_TJ, Country = 'Tajikistan')

# data_18_TJ <- mutate(data_18_TJ, Country = 'TJ')
```

6.1 trying with rbind

The *rbind* command is useful for simple, literal combinations of rows of data (*cbind* is the equivalent to combine columns). However, this method requires that all variables align perfectly between datasets to be combined. We discover some discrepancies when we attempt this.

```
data_2022 <- rbind(data_22_KZ, data_22_KG, data_22_UZ)
```

That generates an error, so let's study the differences between the datasets with *setdiff*. Then use *full_join* to generate a merged dataset that includes all variables. This turns up a problem with incompatible formats in the Year file that will be fixed in the Data cleaning section.

```
setdiff(names(data_22_KG),names(data_22_KZ))
setdiff(names(data_22_KG),names(data_22_UZ))
setdiff(names(data_22_KZ),names(data_22_UZ))

my_test_join <- full_join(data_22_KG, data_22_KZ)
```

7 Data cleaning

As we delve into the data, we discover inconsistencies, such as the formatting of years. Also, date is MM6 in 2018, but labeled 'Year' – gotta watch those questionnaires! We do not care about this level of detail, so let's introduce a similar "Year" variable into the data. This will also overwrite the variable type, but we can double check it. Finally, rename all to 'Year' for consistency.

```
# data_22_KG$MM8
# data_22_KZ$MM8
# data_20_KG$MM8
# data_20_KZ$MM8
# data_18_KG$MM6
# data_18_KZ$MM6

data_22_KG$MM8 <- 2022
data_22_KZ$MM8 <- 2022
data_22_UZ$MM8 <- 2022

data_20_KG$MM8 <- 2020
data_20_KZ$MM8 <- 2020
data_20_UZ$MM8 <- 2020

class(data_22_KG$MM8)
```

```
[1] "numeric"
```

```r
mode(data_22_KG$MM8)
```

```
[1] "numeric"
```

```r
class(data_22_KZ$MM8)
```

```
[1] "numeric"
```

```r
mode(data_22_KZ$MM8)
```

```
[1] "numeric"
```

```r
# data_18_KG$Year
# data_18_KZ$Year

class(data_18_KG$Year)
```

```
[1] "numeric"
```

```r
mode(data_18_KG$Year)
```

```
[1] "numeric"
```

```r
# try again

my_test_join <- full_join(data_22_KG, data_22_KZ)
```

```
Joining with `by = join_by(MM1, MM2b, MM3a, MM3b, MM4, MM5, MM6, MM7, MM8, MM9,
MM10, MM11, MM12, MM13, MM14, A1a, A1b, A2, A3a, A3b, A4a, A4b, A4c, A4d, A4e,
A4f, A5a, A5b, A5c, A5d, A6, A7, B1, B2, B3, B4a, B4b, B5a, B5b, B6a, B6b, B6c,
B6d, B6e, B7, B8, B9a, B9b, B9c, B9d, B10, B11a, B11b, B12, B13, B14, B15a,
B15b, B15c, B15d, B16, D1, D2a, D2b, E1a, E1b, E1c, E1d, E1e, E2a, E2b, E2c,
E2d, E2e, E3, E4a, E4b, E5a, E5b, E5c, E6a, E6b, E7a, E7b, E7c, E8a, E8b, E8c,
E8d, E8e, E9a, E9b, E10, E11, E12, E13, E14, E15a, E15b, E15c, E16, E17, E19a,
E19b, E19c, E20a, E20b, E20c, E21, G1a, G1b, G2, G3a, G3b, G4, G5, G6, G7, H1,
H2, H3, H4, H5a, H5b, DD1, DD2, DD3, DD4, DD5, DD6a, DD6b, DD6c, DD7a, DD7b,
DD7c, DD8, DD9a, DD9b, DD9c, DD9d, DD10, DD11, DD12, DD13, DD14, DD15, DD16,
DD17, DD18a, DD18b, DD19, DD20, EthnicGrp, AgeGrp, FinalWgt1, FinalWgt2,
Country)`
```

```r
# works now

data_22_KG <- data_22_KG |> rename('Year'='MM8')
data_22_KZ <- data_22_KZ |> rename('Year'='MM8')
data_22_UZ <- data_22_UZ |> rename('Year'='MM8')
```

8 Data Wrangling

8.1 Selection of variables

We could merge and work with the entire group of datasets, merging the 9 files together, 3 years x 3 countries. We'd then have access to all the variables. But that is likely to be a lot of work to iron out compatibility. It might be good to do that for a long term project, but we are still just experimenting.

So to take things on a smaller scale, let's just select our variables of interest first. We will pre-clean those and check the variables for compatibility. Then merge the smaller selections, to make the output more comprehensible and browseable.

We look at the smaller 2018 survey first to try to isolate things that are comparable across all years. Questions themselves, their labeling, and naming, does change over time.

In 2018, an interesting question was "what currency do you keep your savings in?" (DD9d), but unfortunately this question did not continue to 2020 and 2022.

We end up looking over the questionnaires quite a bit, but a serious long-term researcher will do even more of that as they try to isolate the most compatible and stable data over time and countries.

Another interesting question is the percentage of income a household receives from remittances (DD9a). Although this question was not asked in 2020, we will capture it to compare 2018 to 2022.

Other standard variables are Gender (DD1), Age (DD2), Marital Status (DD3), Education, (DD4), and Employment Status (DD6a).

There are questions on how favorably residents of a country view the following countries: China (E1a), Russia (E1b), and the USA (E1c).

Also questions on the level of trust in the following news sources: Television (A2a), Newspaper (A2b), Radio (A2c). However this question doesn't continue consistently in later waves, so we use information about the "Main Source of News" (A1a).

Another useful demographic variable is language spoken at home. This variable is coded as DD11 in 2018 and 2022, but DD9 in 2020. This is the type of small fixes that we must undertake in data cleanup.

To capture sentiment, we'll look at if the Country's Financial Situation is getting better or worse (B4a, or B2 in 2018) versus an individual family's situation is getting better or worse (B4b, or B5a in 2018).

Finally, we'll capture the question of whether the Country is going in the Right or Wrong direction (D1).

We'll also need to include the Country and Year variables.

My final list of selected variables DD1 Gender DD2 Age DD3 Marital (Status) DD4 Education DD6a Employment

D1 Direction (Right or Wrong)

E1a China_Favor E1b Russia_Favor E1c USA_Favor

DD9a Remit (Remittances in 2018, 2022)

A1a Main_News (Source)

DD11 Language (at Home) DD9 in 2020

B4a Country_Fin_Sit (B2 in 2018) B4b Family_Fin_Sit (B5a in 2018)

plus keep Country, Year

We will first merge the data files with *full_join*, so that we have one multi-country file for each year. We discover another little problem with the 2018 Uzbek variable HHLandlines that needs reformatting to be consistent with others. Typical data cleaning!

```
mydata2022 <- full_join(data_22_KG, data_22_KZ)
```

```
Joining with `by = join_by(MM1, MM2b, MM3a, MM3b, MM4, MM5, MM6, MM7, Year,
MM9, MM10, MM11, MM12, MM13, MM14, A1a, A1b, A2, A3a, A3b, A4a, A4b, A4c, A4d,
A4e, A4f, A5a, A5b, A5c, A5d, A6, A7, B1, B2, B3, B4a, B4b, B5a, B5b, B6a, B6b,
B6c, B6d, B6e, B7, B8, B9a, B9b, B9c, B9d, B10, B11a, B11b, B12, B13, B14,
B15a, B15b, B15c, B15d, B16, D1, D2a, D2b, E1a, E1b, E1c, E1d, E1e, E2a, E2b,
E2c, E2d, E2e, E3, E4a, E4b, E5a, E5b, E5c, E6a, E6b, E7a, E7b, E7c, E8a, E8b,
E8c, E8d, E8e, E9a, E9b, E10, E11, E12, E13, E14, E15a, E15b, E15c, E16, E17,
E19a, E19b, E19c, E20a, E20b, E20c, E21, G1a, G1b, G2, G3a, G3b, G4, G5, G6,
G7, H1, H2, H3, H4, H5a, H5b, DD1, DD2, DD3, DD4, DD5, DD6a, DD6b, DD6c, DD7a,
DD7b, DD7c, DD8, DD9a, DD9b, DD9c, DD9d, DD10, DD11, DD12, DD13, DD14, DD15,
DD16, DD17, DD18a, DD18b, DD19, DD20, EthnicGrp, AgeGrp, FinalWgt1, FinalWgt2,
Country)`
```

```
mydata2022 <- full_join(mydata2022, data_22_UZ)
```

```
Joining with `by = join_by(MM1, MM2b, MM3a, MM3b, MM4, MM5, MM6, MM7, Year,
MM9, MM10, MM11, MM12, MM13, MM14, A1a, A1b, A2, A3a, A3b, A4a, A4b, A4c, A4d,
A4e, A4f, A5a, A5b, A5c, A5d, A6, A7, B1, B2, B3, B4a, B4b, B5a, B5b, B6a, B6b,
B6c, B6d, B6e, B7, B8, B9a, B9b, B9c, B9d, B10, B11a, B11b, B12, B13, B14,
B15a, B15b, B15c, B15d, B16, D1, D2a, D2b, E1a, E1b, E1c, E1d, E1e, E2a, E2b,
E2c, E2d, E2e, E3, E4a, E4b, E5a, E5b, E5c, E6a, E6b, E7a, E7b, E7c, E8a, E8b,
E8c, E8d, E8e, E9a, E9b, E10, E11, E12, E13, E14, E15a, E15b, E15c, E16, E17,
E19a, E19b, E19c, E20a, E20b, E20c, E21, G1a, G1b, G2, G3a, G3b, G4, G5, G6,
G7, G8, G9, H1, H2, H3, H4, H5a, H5b, DD1, DD2, DD3, DD4, DD5, DD6a, DD6b,
DD6c, DD7a, DD7b, DD7c, DD8, DD9a, DD9b, DD9c, DD9d, DD10, DD11, DD12, DD13,
DD14, DD15, DD16, DD17, DD18a, DD18b, DD19, DD20, EthnicGrp, AgeGrp, FinalWgt1,
FinalWgt2, Country)`
```

```
mydata2020 <- full_join(data_20_KG, data_20_KZ)
```

```
Joining with `by = join_by(MM1, MM2b, MM3a, MM3b, MM4, MM5, MM6, MM7, MM8, MM9,
MM10, MM11, A1a, A1b, A2, A3a, A3b, A4a, A4b, A4c, B1, B2, B3, B4a, B4b, B5a,
B5b, B6a, B6b, B6c, B6d, B6e, B7, B8a, B8b, B8c, B8d, B9, I1, I2, I3, I4, I5a,
I5b, I5c, I5d, I6, I7a, I7b, I7c, I7d, I7e, D1, D2, E1a, E1b, E1c, E1d, E2a,
E2b, E2c, E2d, E3a, E3b, E3c, E4a, E4b, E4c, E5a, E5b, E5c, E6a, E6b, E7a, E7b,
E7c, G1a, G1b, G2, G3, G4, DD1, DD2, DD3, DD4, DD5, DD6a, DD6b, DD6c, DD7, DD8,
DD9, DD10, DD11, DD12, MM12, MM13, MM14, DD13, DD14, DD15, DD16a, DD16b, DD17,
DD18, AgeGrp, EthnicGrp, FinalWgt1, FinalWgt2, Country)`
```

```
mydata2020 <- full_join(mydata2020, data_20_UZ)
```

```
Joining with `by = join_by(MM1, MM2b, MM3a, MM3b, MM4, MM5, MM6, MM7, MM8, MM9,
MM10, MM11, A1a, A1b, A2, A3a, A3b, A4a, A4b, A4c, B1, B2, B3, B4a, B4b, B5a,
B5b, B6a, B6b, B6c, B6d, B6e, B7, B8a, B8b, B8c, B8d, B9, I1, I2, I3, I4, I5a,
I5b, I5c, I5d, I6, I7a, I7b, I7c, I7d, I7e, D1, D2, E1a, E1b, E1c, E1d, E2a,
E2b, E2c, E2d, E3a, E3b, E3c, E4a, E4b, E4c, E5a, E5b, E5c, E6a, E6b, E7a, E7b,
E7c, G1a, G1b, G2, G3, G4, DD1, DD2, DD3, DD4, DD5, DD6a, DD6b, DD6c, DD7, DD8,
DD9, DD10, DD11, DD12, MM12, MM13, MM14, DD13, DD14, DD15, DD16a, DD16b, DD17,
DD18, AgeGrp, EthnicGrp, FinalWgt1, FinalWgt2, Country)`
```

```
mydata2018 <- full_join(data_18_KG, data_18_KZ)
```

```
Joining with `by = join_by(IDNO, Wave, Country, Month, Day, Year, GeoCode,
IntCode, GendInt, SupvCode, Visit, IntBegin, IntEnd, Length, SamPt, NetComp,
NetPhone, Gender, Age, Church, Pray5x, Fast, Work, Occ, Income, IncomeOut,
Homelang, Water, Food, Housing, Electric, Educate, MedCare, Language, PeopPres,
Interact, Comphend, UnEz, BackChk, Method, Format, ThreatHi, Mosque, CelPhon,
OwnPC, Ethnic_M, Region_M, MainNews_M, Topprob_M, TrustGov_M, TrustUS_M,
HrsWorkGrp, AgeGrp, NetSocial, NetChat, Ssite_M, Lastyear_M, nextyear_M,
Statements1, Statements3, CommProb2, OpinionChina, OpinionRussia, OpinionUSA,
OpinionKAZ, OpinionKYR, OpinionTAJ, OpinionUzbek, PMSagintayev, HHLandlines,
NmbrMobilePhones, EighteenPlusMobile, MobileVsLandline, TrustRs_M, LiveHH18,
LLNumber, MobileNumber, HrsWork_M, CanBuy_M, Identfy_M, Religion_M,
ServiceProvider_M, TrustTV, TrustNwsPpr, TrustRad, MostTrust, HaveSatTV,
OpinionTurkey, MarriageStat2, HoH, Occ2, HoHJob, HoHOcc2, PartB4,
FutureSurveys, EducLevel_M, totwt, WatchRoss1, WatchEurAmTV, WatchSputnik,
WatchKhabar, WatchKTK, WatchOTRK, WatchCh5, WatchToj1, WatchSafina, WatchJah,
WatchSMT, WatchRSD, WatchVatan, WatchOzbek, WatchOzbek24, TrustRoss1,
TrustEurAmTV, TrustSputnik, TrustKhabar, TrustKTK, TrustOTRK, TrustCh5,
TrustToj1, TrustSafina, TrustJah, TrustSMT, TrustRSD, TrustOzbek, TrustOzbek24,
MediaEngInfo, GovtCntrlMedia, JournalReport, FairMedia, ForgnInflMedia, TVTime,
NetDomNews, NetIntlNews, Ssite2, SMDomNews, SMIntlNews, SMPoliticians,
SMFrdFam, SMGroups, EEUBenefit, HighPrices, CostFood, CostElect, CostBus,
CostHouse, Statements5, LastYrRemit, LLCalls, MobCalls, ServiceProvider2,
TypeOcc_M, HoHTypeOcc_M, WatchAA, WatchTV4, WatchOvaz, WatchNeit, WatchTurkPM,
OwnCar, AccessFood, OpinionTurkm, EconRelChina, EconRelRus, EconRelUS,
PMAbylgaziev, PlcHld, AgeGroup2, WhyRight1_M, WhyRight2_M, WhyWrong1_M,
WhyWrong2_M, WhyOpRus1_M, WhyOpRus2_M, WhyOpUS1_M, WhyOpUS2_M, BankAcc_M,
```

```
KeepMoney_M, MainIntNews_M, TVType_M, EconNY_M, BuyFood_M, RiteRong_M,
Improve1_M, Improve2_M, ChinaEng_M, ChinaEngWor_M, Party_M, TodayEcSit_M)`
```

```r
# mydata2018 <- full_join(mydata2018, data_18_UZ)

# a problem with $$HHLandlines in last line
data_18_UZ$HHLandlines <- as.character(data_18_UZ$HHLandlines)
mydata2018 <- full_join(mydata2018, data_18_UZ)
```

```
Joining with `by = join_by(IDNO, Wave, Country, Month, Day, Year, GeoCode,
IntCode, GendInt, SupvCode, Visit, IntBegin, IntEnd, Length, SamPt, NetComp,
NetPhone, Gender, Age, Church, Pray5x, Fast, Work, Occ, Income, IncomeOut,
Homelang, Water, Food, Housing, Electric, Educate, MedCare, Language, PeopPres,
Interact, Comphend, UnEz, BackChk, Method, Format, ThreatHi, Mosque, CelPhon,
OwnPC, Ethnic_M, Region_M, MainNews_M, Topprob_M, TrustGov_M, TrustUS_M,
HrsWorkGrp, AgeGrp, NetSocial, NetChat, Ssite_M, Lastyear_M, nextyear_M,
Statements1, Statements3, CommProb2, OpinionChina, OpinionRussia, OpinionUSA,
OpinionKAZ, OpinionKYR, OpinionTAJ, OpinionUzbek, PMSagintayev, HHLandlines,
NmbrMobilePhones, EighteenPlusMobile, MobileVsLandline, TrustRs_M, LiveHH18,
LLNumber, MobileNumber, HrsWork_M, CanBuy_M, Identfy_M, Religion_M,
ServiceProvider_M, TrustTV, TrustNwsPpr, TrustRad, MostTrust, HaveSatTV,
OpinionTurkey, MarriageStat2, HoH, Occ2, HoHJob, HoHOcc2, PartB4,
FutureSurveys, EducLevel_M, totwt, WatchRoss1, WatchEurAmTV, WatchSputnik,
WatchKhabar, WatchKTK, WatchOTRK, WatchCh5, WatchToj1, WatchSafina, WatchJah,
WatchSMT, WatchRSD, WatchVatan, WatchOzbek, WatchOzbek24, TrustRoss1,
TrustEurAmTV, TrustSputnik, TrustKhabar, TrustKTK, TrustOTRK, TrustCh5,
TrustToj1, TrustSafina, TrustJah, TrustSMT, TrustRSD, TrustOzbek, TrustOzbek24,
MediaEngInfo, GovtCntrlMedia, JournalReport, FairMedia, ForgnInflMedia, TVTime,
NetDomNews, NetIntlNews, Ssite2, SMDomNews, SMIntlNews, SMPoliticians,
SMFrdFam, SMGroups, EEUBenefit, HighPrices, CostFood, CostElect, CostBus,
CostHouse, Statements5, LastYrRemit, LLCalls, MobCalls, ServiceProvider2,
TypeOcc_M, HoHTypeOcc_M, WatchAA, WatchTV4, WatchOvaz, WatchNeit, WatchTurkPM,
OwnCar, AccessFood, OpinionTurkm, EconRelChina, EconRelRus, EconRelUS,
PMAbylgaziev, PlcHld, AgeGroup2, WhyRight1_M, WhyRight2_M, WhyWrong1_M,
WhyWrong2_M, WhyOpRus1_M, WhyOpRus2_M, WhyOpUS1_M, WhyOpUS2_M, BankAcc_M,
KeepMoney_M, MainIntNews_M, TVType_M, EconNY_M, BuyFood_M, RiteRong_M,
Improve1_M, Improve2_M, ChinaEng_M, ChinaEngWor_M, Party_M, TodayEcSit_M)`
```

Check the names and then rename them to friendlier text names. Then select the list of variables of interest. There are few extra little twists in the code below, driven by quirks in the data. Especially some naming surprises in the 2018 data, since they already use text names, just differently and undocumented!

```r
names(mydata2018)
```

```
  [1] "IDNO"                "Wave"                "Country"
  [4] "Month"               "Day"                 "Year"
  [7] "GeoCode"             "IntCode"             "GendInt"
```

```
 [10] "SupvCode"           "Visit"              "IntBegin"
 [13] "IntEnd"             "Length"             "SamPt"
 [16] "NetComp"            "NetPhone"           "Gender"
 [19] "Age"                "Church"             "Pray5x"
 [22] "Fast"               "Work"               "Occ"
 [25] "Income"             "IncomeOut"          "Homelang"
 [28] "Water"              "Food"               "Housing"
 [31] "Electric"           "Educate"            "MedCare"
 [34] "Language"           "PeopPres"           "Interact"
 [37] "Comphend"           "UnEz"               "BackChk"
 [40] "Method"             "Format"             "ThreatHi"
 [43] "Mosque"             "CelPhon"            "OwnPC"
 [46] "Ethnic_M"           "Region_M"           "MainNews_M"
 [49] "Topprob_M"          "TrustGov_M"         "TrustUS_M"
 [52] "HrsWorkGrp"         "AgeGrp"             "NetSocial"
 [55] "NetChat"            "Ssite_M"            "Lastyear_M"
 [58] "nextyear_M"         "Statements1"        "Statements3"
 [61] "CommProb2"          "OpinionChina"       "OpinionRussia"
 [64] "OpinionUSA"         "OpinionKAZ"         "OpinionKYR"
 [67] "OpinionTAJ"         "OpinionUzbek"       "PMSagintayev"
 [70] "HHLandlines"        "NmbrMobilePhones"   "EighteenPlusMobile"
 [73] "MobileVsLandline"   "TrustRs_M"          "LiveHH18"
 [76] "LLNumber"           "MobileNumber"       "HrsWork_M"
 [79] "CanBuy_M"           "Identfy_M"          "Religion_M"
 [82] "ServiceProvider_M"  "TrustTV"            "TrustNwsPpr"
 [85] "TrustRad"           "MostTrust"          "HaveSatTV"
 [88] "OpinionTurkey"      "MarriageStat2"      "HoH"
 [91] "Occ2"               "HoHJob"             "HoHOcc2"
 [94] "PartB4"             "FutureSurveys"      "EducLevel_M"
 [97] "totwt"              "WatchRoss1"         "WatchEurAmTV"
[100] "WatchSputnik"       "WatchKhabar"        "WatchKTK"
[103] "WatchOTRK"          "WatchCh5"           "WatchToj1"
[106] "WatchSafina"        "WatchJah"           "WatchSMT"
[109] "WatchRSD"           "WatchVatan"         "WatchOzbek"
[112] "WatchOzbek24"       "TrustRoss1"         "TrustEurAmTV"
[115] "TrustSputnik"       "TrustKhabar"        "TrustKTK"
[118] "TrustOTRK"          "TrustCh5"           "TrustToj1"
[121] "TrustSafina"        "TrustJah"           "TrustSMT"
[124] "TrustRSD"           "TrustOzbek"         "TrustOzbek24"
[127] "MediaEngInfo"       "GovtCntrlMedia"     "JournalReport"
[130] "FairMedia"          "ForgnInflMedia"     "TVTime"
[133] "NetDomNews"         "NetIntlNews"        "Ssite2"
[136] "SMDomNews"          "SMIntlNews"         "SMPoliticians"
[139] "SMFrdFam"           "SMGroups"           "EEUBenefit"
[142] "HighPrices"         "CostFood"           "CostElect"
[145] "CostBus"            "CostHouse"          "Statements5"
[148] "LastYrRemit"        "LLCalls"            "MobCalls"
[151] "ServiceProvider2"   "TypeOcc_M"          "HoHTypeOcc_M"
```

```
[154] "WatchAA"            "WatchTV4"           "WatchOvaz"
[157] "WatchNeit"          "WatchTurkPM"        "OwnCar"
[160] "AccessFood"         "OpinionTurkm"       "EconRelChina"
[163] "EconRelRus"         "EconRelUS"          "PMAbylgaziev"
[166] "PlcHld"             "AgeGroup2"          "WhyRight1_M"
[169] "WhyRight2_M"        "WhyWrong1_M"        "WhyWrong2_M"
[172] "WhyOpRus1_M"        "WhyOpRus2_M"        "WhyOpUS1_M"
[175] "WhyOpUS2_M"         "BankAcc_M"          "KeepMoney_M"
[178] "MainIntNews_M"      "TVType_M"           "EconNY_M"
[181] "BuyFood_M"          "RiteRong_M"         "Improve1_M"
[184] "Improve2_M"         "ChinaEng_M"         "ChinaEngWor_M"
[187] "Party_M"            "TodayEcSit_M"
```

```
names(mydata2020)
```

```
  [1] "MM1"       "MM2b"      "MM3a"      "MM3b"      "MM4"       "MM5"
  [7] "MM6"       "MM7"       "MM8"       "MM9"       "MM10"      "MM11"
 [13] "A1a"       "A1b"       "A2"        "A3a"       "A3b"       "A4a"
 [19] "A4b"       "A4c"       "B1"        "B2"        "B3"        "B4a"
 [25] "B4b"       "B5a"       "B5b"       "B6a"       "B6b"       "B6c"
 [31] "B6d"       "B6e"       "B7"        "B8a"       "B8b"       "B8c"
 [37] "B8d"       "B9"        "I1"        "I2"        "I3"        "I4"
 [43] "I5a"       "I5b"       "I5c"       "I5d"       "I6"        "I7a"
 [49] "I7b"       "I7c"       "I7d"       "I7e"       "D1"        "D2"
 [55] "E1a"       "E1b"       "E1c"       "E1d"       "E2a"       "E2b"
 [61] "E2c"       "E2d"       "E3a"       "E3b"       "E3c"       "E4a"
 [67] "E4b"       "E4c"       "E5a"       "E5b"       "E5c"       "E6a"
 [73] "E6b"       "E7a"       "E7b"       "E7c"       "G1a"       "G1b"
 [79] "G2"        "G3"        "G4"        "DD1"       "DD2"       "DD3"
 [85] "DD4"       "DD5"       "DD6a"      "DD6b"      "DD6c"      "DD7"
 [91] "DD8"       "DD9"       "DD10"      "DD11"      "DD12"      "MM12"
 [97] "MM13"      "MM14"      "DD13"      "DD14"      "DD15"      "DD16a"
[103] "DD16b"     "DD17"      "DD18"      "AgeGrp"    "EthnicGrp" "FinalWgt1"
[109] "FinalWgt2" "Country"
```

```
names(mydata2022)
```

```
  [1] "MM1"       "MM2b"      "MM3a"      "MM3b"      "MM4"       "MM5"
  [7] "MM6"       "MM7"       "Year"      "MM9"       "MM10"      "MM11"
 [13] "MM12"      "MM13"      "MM14"      "A1a"       "A1b"       "A2"
 [19] "A3a"       "A3b"       "A4a"       "A4b"       "A4c"       "A4d"
 [25] "A4e"       "A4f"       "A5a"       "A5b"       "A5c"       "A5d"
 [31] "A6"        "A7"        "B1"        "B2"        "B3"        "B4a"
 [37] "B4b"       "B5a"       "B5b"       "B6a"       "B6b"       "B6c"
 [43] "B6d"       "B6e"       "B7"        "B8"        "B9a"       "B9b"
 [49] "B9c"       "B9d"       "B10"       "B11a"      "B11b"      "B12"
 [55] "B13"       "B14"       "B15a"      "B15b"      "B15c"      "B15d"
 [61] "B16"       "D1"        "D2a"       "D2b"       "E1a"       "E1b"
```

```
 [67] "E1c"       "E1d"       "E1e"       "E2a"       "E2b"       "E2c"
 [73] "E2d"       "E2e"       "E3"        "E4a"       "E4b"       "E5a"
 [79] "E5b"       "E5c"       "E6a"       "E6b"       "E7a"       "E7b"
 [85] "E7c"       "E8a"       "E8b"       "E8c"       "E8d"       "E8e"
 [91] "E9a"       "E9b"       "E10"       "E11"       "E12"       "E13"
 [97] "E14"       "E15a"      "E15b"      "E15c"      "E16"       "E17"
[103] "E19a"      "E19b"      "E19c"      "E20a"      "E20b"      "E20c"
[109] "E21"       "G1a"       "G1b"       "G2"        "G3a"       "G3b"
[115] "G4"        "G5"        "G6"        "G7"        "G8"        "G9"
[121] "H1"        "H2"        "H3"        "H4"        "H5a"       "H5b"
[127] "DD1"       "DD2"       "DD3"       "DD4"       "DD5"       "DD6a"
[133] "DD6b"      "DD6c"      "DD7a"      "DD7b"      "DD7c"      "DD8"
[139] "DD9a"      "DD9b"      "DD9c"      "DD9d"      "DD10"      "DD11"
[145] "DD12"      "DD13"      "DD14"      "DD15"      "DD16"      "DD17"
[151] "DD18a"     "DD18b"     "DD19"      "DD20"      "EthnicGrp" "AgeGrp"
[157] "FinalWgt1" "FinalWgt2" "Country"   "MM1a"      "G8_1"      "G9_1"
[163] "G8_2"      "G9_2"
```

```
mydata2022 <- mydata2022 |> rename('Sex'='DD1', 'Age'='DD2', 'Marital'='DD3',
                                   'Education'='DD4', 'Employment'='DD6a',
                                   'Direction'='D1', 'China_Favor'='E1a',
                                   'Russia_Favor'='E1b','USA_Favor'='E1c',
                                   'Main_News'='A1a', 'Remit'='DD9a',
                                   'Language'='DD11','Country_Fin_Sit'='B4a',
                                   'Family_Fin_Sit'='B4b')

mydata2022_select <- mydata2022 |> select('Sex', 'Age', 'Marital',
 ↪  'Education',
                                          'Employment', 'Direction',
                                           ↪  'China_Favor',
                                          'Russia_Favor','USA_Favor',
                                           ↪  'Main_News',
                                          'Remit',
                                           ↪  'Language','Country_Fin_Sit',
                                          'Family_Fin_Sit', 'Country',
                                           ↪  'Year')

# note we rename Year here from MM8

mydata2020 <- mydata2020 |> rename('Sex'='DD1', 'Age'='DD2', 'Marital'='DD3',
                                   'Education'='DD4', 'Employment'='DD6a',
                                   'Direction'='D1', 'China_Favor'='E1a',
                                   'Russia_Favor'='E1b','USA_Favor'='E1c',
                                   'Main_News'='A1a', 'Language'='DD9',
                                   'Country_Fin_Sit'='B4a',
                                    ↪  'Family_Fin_Sit'='B4b',
                                   'Year'='MM8')
```

```
mydata2020_select <- mydata2020 |> select('Sex', 'Age', 'Marital',
 ↪  'Education',
                                          'Employment', 'Direction',
                                           ↪  'China_Favor',
                                          'Russia_Favor','USA_Favor',
                                           ↪  'Main_News',
                                          'Language','Country_Fin_Sit',
                                          'Family_Fin_Sit', 'Country',
                                           ↪  'Year')

# need to add a blank 'Remit' variable here

mydata2020_select <- mydata2020_select |> mutate(Remit=NA)

# and 2018 data already uses the names, but formatted differently
# codebook is not satisfactory, does not describe these labels, have to judge
 ↪  by answers in some cases
# note the 'LangExtra' to avoid repeating a name

mydata2018 <- mydata2018 |> rename('Sex'='Gender', 'Marital'='MarriageStat2',
                                  'Education'='EducLevel_M',
                                   ↪  'Employment'='Work',
                                  'Direction'='RiteRong_M',
                                   ↪  'China_Favor'='OpinionChina',

                                   ↪  'Russia_Favor'='OpinionRussia','USA_Favor'='OpinionUSA'
                                  'Main_News'='MainNews_M',
                                   ↪  'Language'='Homelang','LangExtra'='Language',
                                  'Country_Fin_Sit'='TodayEcSit_M',
                                   ↪  'Family_Fin_Sit'='Lastyear_M',
                                  'Remit'='IncomeOut')

mydata2018_select <- mydata2018 |> select('Sex', 'Age', 'Marital',
 ↪  'Education',
                                          'Employment', 'Direction',
                                           ↪  'China_Favor',
                                          'Russia_Favor','USA_Favor',
                                           ↪  'Main_News',
                                          'Language','Country_Fin_Sit',
                                           ↪  'Remit',
                                          'Family_Fin_Sit', 'Country',
                                           ↪  'Year')
```

We could try more full joins, but since we have already manually reset the variable names to be consistent across all three years, here we can simply to a direct *rbind* to paste the three files together, and then double check that it came out as expected.

```
my_final_data <- rbind(mydata2018_select, mydata2020_select,
 ↪  mydata2022_select)
```

```r
str(my_final_data)
```

```
tibble [15,029 x 16] (S3: tbl_df/tbl/data.frame)
 $ Sex            : chr [1:15029] "Female" "Female" "Female" "Female" ...
 $ Age            : num [1:15029] 29 83 48 37 43 60 34 42 18 55 ...
 $ Marital        : chr [1:15029] "Single" "Widowed or divorced" "Single" "Married" ...
 $ Education      : chr [1:15029] "Complete higher education" "Complete higher education" "Voc
 $ Employment     : chr [1:15029] "Unemployed and actively seeking employment" "Retired" "Unem
time" ...
 $ Direction      : chr [1:15029] "Right direction" "Wrong direction" "Don't Know (vol.)" "Wro
 $ China_Favor    : chr [1:15029] "Very Unfavorable" "Somewhat Favorable" "Somewhat Favorable"
 $ Russia_Favor   : chr [1:15029] "Very Favorable" "Very Favorable" "Somewhat Favorable" "Don'
 $ USA_Favor      : chr [1:15029] "Somewhat Unfavorable" "Don't Know (vol.)" "Somewhat Unfavor
 $ Main_News      : chr [1:15029] "Internet" "Russian television stations" "Russian television
 $ Language       : chr [1:15029] "Russian" "Russian" "Russian" "Russian" ...
 $ Country_Fin_Sit: chr [1:15029] "About the Same" "About the Same" "Don't Know (vol.)" "About
 $ Remit          : chr [1:15029] "Not Applicable / No Relatives Abroad / No Remittances from
 $ Family_Fin_Sit : chr [1:15029] "Become Worse" "Become Worse" "Become Worse" "Stayed the Sam
 $ Country        : chr [1:15029] "Kyrgyzstan" "Kyrgyzstan" "Kyrgyzstan" "Kyrgyzstan" ...
 $ Year           : num [1:15029] 2018 2018 2018 2018 2018 ...
```

```r
summary(my_final_data)
```

```
     Sex                 Age          Marital           Education
 Length:15029       Min.   :18.00   Length:15029       Length:15029
 Class :character   1st Qu.:27.00   Class :character   Class :character
 Mode  :character   Median :35.00   Mode  :character   Mode  :character
                    Mean   :37.92
                    3rd Qu.:47.00
                    Max.   :98.00
  Employment         Direction         China_Favor        Russia_Favor
 Length:15029       Length:15029       Length:15029       Length:15029
 Class :character   Class :character   Class :character   Class :character
 Mode  :character   Mode  :character   Mode  :character   Mode  :character



  USA_Favor          Main_News           Language         Country_Fin_Sit
 Length:15029       Length:15029       Length:15029       Length:15029
 Class :character   Class :character   Class :character   Class :character
 Mode  :character   Mode  :character   Mode  :character   Mode  :character



    Remit           Family_Fin_Sit       Country               Year
 Length:15029       Length:15029       Length:15029       Min.   :2018
 Class :character   Class :character   Class :character   1st Qu.:2018
 Mode  :character   Mode  :character   Mode  :character   Median :2020
```

```
                                                                    Mean   :2020
                                                                    3rd Qu.:2022
                                                                    Max.   :2022
```

```
attach(my_final_data)
```

8.2 Descriptive statistics

Let's take a look at several of these variables and see how they're distributed. A simple *table* command will do the trick for getting initial familiarity with the data. While some large datasets are difficult to gain intuitive understanding of, for this kind of social science data, nothing beats spending time looking over the data before getting into more complex analysis. We may also discover outliers or issues to remedy before moving forward. We discover an issue with the Remittance (Remit) variable that needs to be coerced into numeric form.

We can run any cross-tabs by listing two variables in the *table* command. Many combinations can be explored (not shown here).

```
table(Sex)
```

```
Sex
Female   Male
  7501   7528
```

```
table(Age)
```

```
Age
 18  19  20  21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36  37
431 314 406 349 373 353 376 501 386 499 465 391 669 379 530 418 366 473 372 356
 38  39  40  41  42  43  44  45  46  47  48  49  50  51  52  53  54  55  56  57
376 266 455 223 309 289 203 335 240 222 249 198 341 153 209 212 168 200 147 144
 58  59  60  61  62  63  64  65  66  67  68  69  70  71  72  73  74  75  76  77
170 119 242 128 113 139  89 123  65  56  49  45  89  33  41  34  15  21  17  13
 78  79  80  81  82  83  84  85  87  88  89  90  93  95  98
 14   8  19  10   7   4   4   3   2   2   1   1   1   1   5
```

```
table(Marital)
```

```
Marital
  Don't Know (vol.)             Married      Refused (vol.)              Single
                  9               10072                  53                3266
Widowed or divorced
               1629
```

```
table(Education)
```

```
Education
                            Complete higher education
                                                 3825
                                       Complete school
                                                 3161
Complete vocational - vocational school, lyceum, etc.
                                                 3176
                                     Don't Know (vol.)
                                                    4
                                       Finished school
                                                 1510
                           Incomplete higher education
                                                 1039
                                     Incomplete school
                                                  382
                       Incomplete vocational education
                                                  524
                                          No education
                                                   25
                                        Refused (vol.)
                                                   12
          Vocational (vocational school, lyceum etc.)
                                                 1371
```

```
table(Employment)
```

```
Employment
                                    A housewife
                                            961
                     A housewife / househusband
                                           1872
                                      A student
                                            809
                                Don't Know (vol.)
                                             20
                                   Refused (vol.)
                                             23
                                        Retired
                                           1611
    Unemployed and actively seeking employment
                                           1193
Unemployed and not actively seeking employment
                                            668
                                Working full-time
                                           5842
                                Working part-time
                                           2030
```

```
table(Direction)
```

```
Direction
Don't Know (vol.)         Not asked    Refused (vol.)   Right direction
             1563              1500               223              9885
  Wrong direction
             1858
```

```
table(China_Favor)
```

```
China_Favor
   Don't Know (vol.)       Refused (vol.)   Somewhat Favorable
                1986                  161                 5736
Somewhat Unfavorable       Very Favorable     Very Unfavorable
                2739                 1811                 2596
```

```
table(Sex)
```

```
Sex
Female   Male
  7501   7528
```

```
table(Russia_Favor)
```

```
Russia_Favor
   Don't Know (vol.)       Refused (vol.)   Somewhat Favorable
                1116                  105                 7439
Somewhat Unfavorable       Very Favorable     Very Unfavorable
                1367                 4282                  720
```

```
table(USA_Favor)
```

```
USA_Favor
   Don't Know (vol.)       Refused (vol.)   Somewhat Favorable
                3213                  193                 4734
Somewhat Unfavorable       Very Favorable     Very Unfavorable
                2714                 2113                 2062
```

```
table(Main_News)
```

```
Main_News
                                 Don't Know (vol.)
                                               333
                     Family, friends, or neighbors
                                               313
                                          Internet
                                              5397
                           Internet: blogs, others
                                               614
                                        Local radio
```

```
                                                 33
Non-government newspapers published in our country
                                                 11
              None / Not interested in news (vol.)
                                                337
                                       Other (vol.)
                                                 52
                          Other foreign news sources
                                                 11
                           Other television stations
                                                 11
        Our country's national television stations
                                               5526
                        Our government's newspapers
                                                 71
                                              Radio
                                                 40
                                     Refused (vol.)
                                                 23
                  Religious services or literature
                                                 10
                                 Russian newspapers
                                                  5
                                       Russian radio
                                                 13
                        Russian television stations
                                                263
                  Social media sites or messengers
                                               1966
```

```
table(Country_Fin_Sit)
```

```
Country_Fin_Sit
   About the Same Don't Know (vol.)       Much Better        Much Worse
             2790               555              3292              1745
   Refused (vol.)   Somewhat Better    Somewhat Worse
               73              4188              2386
```

```
table(Remit)
```

```
Remit
                                                                  1
                                                                  6
                                                 1.00000000000000000
                                                                  4
                                                                 10
                                                                 64
                                                10.00000000000000000
```

28
100
130
100.00000000000000000
29
14.00000000000000000
1
15
17
15.00000000000000000
7
17
1
2
6
2.00000000000000000
4
20
60
20.00000000000000000
19
24.00000000000000000
1
25
7
25.00000000000000000
5
3
5
3.00000000000000000
3
30
80
30.00000000000000000
26
33
1
33.00000000000000000
1
35
4
35.00000000000000000
1
4
3
4.00000000000000000
3
40

41
40.00000000000000000
16
45
1
5
15
5.00000000000000000
14
50
218
50.00000000000000000
62
6.00000000000000000
1
60
33
60.00000000000000000
9
65.00000000000000000
1
7
2
70
84
70.00000000000000000
18
75
7
75.00000000000000000
1
8
1
8.00000000000000000
1
80
61
80.00000000000000000
12
85
5
9.00000000000000000
1
90
50
90.00000000000000000
15
95

```
                                                                    2
                                                 95.00000000000000000
                                                                    1
                                                                   98
                                                                    2
                                                                   99
                                                                    2
                                                     Don't Know (vol.)
                                                                  310
            Not Applicable / No Relatives Abroad / No Remittances from A
                                                                 3628
Not Applicable / No Relatives Abroad / No Remittances from Abroad (vol.)
                                                                 3745
                                                        Refused (vol.)
                                                                  154
```

```
# fix the problem with Remit

my_final_data$Remit<-as.numeric(Remit)
```

```
Warning: NAs introduced by coercion
```

```
attach(my_final_data)
```

```
The following objects are masked from my_final_data (pos = 3):

    Age, China_Favor, Country, Country_Fin_Sit, Direction, Education,
    Employment, Family_Fin_Sit, Language, Main_News, Marital, Remit,
    Russia_Favor, Sex, USA_Favor, Year
```

```
table(Remit)
```

```
Remit
  1   2   3   4   5   6   7   8   9  10  14  15  17  20  24  25  30  33  35  40
 10  10   8   6  29   1   2   2   1  92   1  24   1  79   1  12 106   2   5  57
 45  50  60  65  70  75  80  85  90  95  98  99 100
  1 280  42   1 102   8  73   5  65   3   2   2 159
```

```
table(Family_Fin_Sit)
```

```
Family_Fin_Sit
   About the same     Become Better      Become Worse Don't Know (vol.)
             3658              2446               453               148
      Much better       Much Better        Much worse        Much Worse
             1041               705               255               611
   Refused (vol.)   Somewhat better   Somewhat Better    Somewhat worse
               50              1371              1323               427
   Somewhat Worse   Stayed the Same
              987              1554
```

```r
table(Country)
```

```
Country
Kazakhstan Kyrgyzstan Uzbekistan
      5011       5018       5000
```

```r
table(Year)
```

```
Year
2018 2019 2020 2022
4473   27 6000 4529
```

```r
table(Direction, Sex)
```

```
                   Sex
Direction           Female Male
  Don't Know (vol.)    883  680
  Not asked            629  871
  Refused (vol.)       101  122
  Right direction     4993 4892
  Wrong direction      895  963
```

```r
table(Marital, Employment)
```

```
                     Employment
Marital               A housewife A housewife / househusband A student
  Don't Know (vol.)             1                          1         0
  Married                     834                       1616        80
  Refused (vol.)                5                          2         2
  Single                       54                        136       720
  Widowed or divorced          67                        117         7
                     Employment
Marital               Don't Know (vol.) Refused (vol.) Retired
  Don't Know (vol.)                   0              0       0
  Married                            16              7     988
  Refused (vol.)                      0             10       3
  Single                              2              5      65
  Widowed or divorced                 2              1     555
                     Employment
Marital               Unemployed and actively seeking employment
  Don't Know (vol.)                                            2
  Married                                                    777
  Refused (vol.)                                               4
  Single                                                     318
  Widowed or divorced                                         92
                     Employment
Marital               Unemployed and not actively seeking employment
  Don't Know (vol.)                                                0
```

```
  Married                                                        446
  Refused (vol.)                                                   3
  Single                                                         175
  Widowed or divorced                                             44
                     Employment
Marital               Working full-time Working part-time
  Don't Know (vol.)                   4                 1
  Married                          3980              1328
  Refused (vol.)                     15                 9
  Single                           1299               492
  Widowed or divorced               544               200
```

```
table(Marital, Education)
```

```
                     Education
Marital               Complete higher education Complete school
  Don't Know (vol.)                           1               2
  Married                                  2765            2285
  Refused (vol.)                             22               2
  Single                                    645             564
  Widowed or divorced                       392             308
                     Education
Marital               Complete vocational - vocational school, lyceum, etc.
  Don't Know (vol.)                                                      1
  Married                                                             2114
  Refused (vol.)                                                         4
  Single                                                               767
  Widowed or divorced                                                  290
                     Education
Marital               Don't Know (vol.) Finished school
  Don't Know (vol.)                   1               1
  Married                             1            1101
  Refused (vol.)                      0               1
  Single                              1             177
  Widowed or divorced                 1             230
                     Education
Marital               Incomplete higher education Incomplete school
  Don't Know (vol.)                             0                 0
  Married                                     367               251
  Refused (vol.)                                3                 0
  Single                                      619                63
  Widowed or divorced                          50                68
                     Education
Marital               Incomplete vocational education No education
  Don't Know (vol.)                                 2            0
  Married                                         304           14
  Refused (vol.)                                    2            1
  Single                                          151            3
```

```
  Widowed or divorced                                65            7
                     Education
Marital               Refused (vol.)
  Don't Know (vol.)                0
  Married                          0
  Refused (vol.)                   9
  Single                           3
  Widowed or divorced              0
                     Education
Marital               Vocational (vocational school, lyceum etc.)
  Don't Know (vol.)                                                1
  Married                                                        870
  Refused (vol.)                                                   9
  Single                                                         273
  Widowed or divorced                                            218
```

```
table(Education, Employment)
```

```
                                                        Employment
Education                                                A housewife
  Complete higher education                                      127
  Complete school                                                  0
  Complete vocational - vocational school, lyceum, etc.            0
  Don't Know (vol.)                                                0
  Finished school                                                458
  Incomplete higher education                                     19
  Incomplete school                                               25
  Incomplete vocational education                                 62
  No education                                                     0
  Refused (vol.)                                                   0
  Vocational (vocational school, lyceum etc.)                    270
                                                        Employment
Education                                                A housewife / househusband
  Complete higher education                                                    327
  Complete school                                                              815
  Complete vocational - vocational school, lyceum, etc.                        541
  Don't Know (vol.)                                                              1
  Finished school                                                                0
  Incomplete higher education                                                   56
  Incomplete school                                                             77
  Incomplete vocational education                                               47
  No education                                                                   8
  Refused (vol.)                                                                 0
  Vocational (vocational school, lyceum etc.)                                    0
                                                        Employment
Education                                                A student
  Complete higher education                                       22
  Complete school                                                107
```

Complete vocational - vocational school, lyceum, etc.	76
Don't Know (vol.)	0
Finished school	43
Incomplete higher education	440
Incomplete school	13
Incomplete vocational education	85
No education	0
Refused (vol.)	0
Vocational (vocational school, lyceum etc.)	23

Education	Employment Don't Know (vol.)
Complete higher education	8
Complete school	6
Complete vocational - vocational school, lyceum, etc.	4
Don't Know (vol.)	1
Finished school	0
Incomplete higher education	1
Incomplete school	0
Incomplete vocational education	0
No education	0
Refused (vol.)	0
Vocational (vocational school, lyceum etc.)	0

Education	Employment Refused (vol.)	Retired
Complete higher education	4	355
Complete school	3	256
Complete vocational - vocational school, lyceum, etc.	0	238
Don't Know (vol.)	1	0
Finished school	3	322
Incomplete higher education	1	26
Incomplete school	1	80
Incomplete vocational education	0	68
No education	0	2
Refused (vol.)	8	0
Vocational (vocational school, lyceum etc.)	2	264

Education	Employment Unemployed and actively seeking employ
Complete higher education	
Complete school	
Complete vocational - vocational school, lyceum, etc.	
Don't Know (vol.)	
Finished school	
Incomplete higher education	
Incomplete school	
Incomplete vocational education	
No education	
Refused (vol.)	
Vocational (vocational school, lyceum etc.)	

```
                                                      Employment
Education                                              Unemployed and not actively seeking em
  Complete higher education
  Complete school
  Complete vocational - vocational school, lyceum, etc.
  Don't Know (vol.)
  Finished school
  Incomplete higher education
  Incomplete school
  Incomplete vocational education
  No education
  Refused (vol.)
  Vocational (vocational school, lyceum etc.)
                                                      Employment
Education                                              Working full-time
  Complete higher education                                         2240
  Complete school                                                   1008
  Complete vocational - vocational school, lyceum, etc.             1351
  Don't Know (vol.)                                                    1
  Finished school                                                    269
  Incomplete higher education                                        275
  Incomplete school                                                   85
  Incomplete vocational education                                    133
  No education                                                         6
  Refused (vol.)                                                       1
  Vocational (vocational school, lyceum etc.)                        473
                                                      Employment
Education                                              Working part-time
  Complete higher education                                         435
  Complete school                                                   502
  Complete vocational - vocational school, lyceum, etc.             493
  Don't Know (vol.)                                                    0
  Finished school                                                    169
  Incomplete higher education                                        153
  Incomplete school                                                   53
  Incomplete vocational education                                     57
  No education                                                         2
  Refused (vol.)                                                       2
  Vocational (vocational school, lyceum etc.)                        164
```

8.3 Recoding

We notice some issues that we also want to clean up to simplify the presentation of data, such as recoding to collapse the "Don't Know" and "Refused" responses into one NA (not available) label. This will help us focus the visualizations on the meaningful categories.

Let's also do some more extensive recoding to simplify some of the multiple categories of semi-inconsistent responses that we see here.

```r
my_final_data$Marital_recode <- replace_values(
  Marital,
  c("Don't Know (vol.)", "Refused (vol.)") ~ NA
)

my_final_data$Employment_recode <- replace_values(
  Employment,
  c("Don't Know (vol.)", "Refused (vol.)") ~ NA
)

my_final_data$Employment_recode2 <- replace_values(
  Employment,
  c("Working part-time", "Working full-time") ~ "Working",
  c("Unemployed and not actively seeking employment", "Unemployed and
   ↪  actively seeking employment") ~ "Unemployed",
  c("A housewife / househusband","A housewife") ~ "Housewife",
  c("Don't Know (vol.)", "Refused (vol.)") ~ NA
)

my_final_data$Direction_recode <- replace_values(
  Direction,
  c("Don't Know (vol.)", "Refused (vol.)", "Not asked") ~ NA
)

my_final_data$Family_Fin_Sit_recode <- replace_values(
  Family_Fin_Sit,
  c("Much better", "Much Better", "Become Better", "Somewhat better",
   ↪  "Somewhat Better") ~ "Better",
  c("About the same", "Stayed the Same") ~ "Same",
  c("Become Worse","Much Worse", "Much worse", "Somewhat Worse", "Somewhat
   ↪  worse") ~ "Worse",
  c("Don't Know (vol.)", "Refused (vol.)") ~ NA
)

attach(my_final_data)

The following objects are masked from my_final_data (pos = 3):

    Age, China_Favor, Country, Country_Fin_Sit, Direction, Education,
    Employment, Family_Fin_Sit, Language, Main_News, Marital, Remit,
    Russia_Favor, Sex, USA_Favor, Year

The following objects are masked from my_final_data (pos = 4):

    Age, China_Favor, Country, Country_Fin_Sit, Direction, Education,
    Employment, Family_Fin_Sit, Language, Main_News, Marital, Remit,
    Russia_Favor, Sex, USA_Favor, Year

table(Marital_recode,Employment_recode)
```

```
                     Employment_recode
Marital_recode        A housewife A housewife / househusband A student Retired
  Married                    834                       1616        80     988
  Single                      54                        136       720      65
  Widowed or divorced         67                        117         7     555
                     Employment_recode
Marital_recode        Unemployed and actively seeking employment
  Married                                                   777
  Single                                                    318
  Widowed or divorced                                        92
                     Employment_recode
Marital_recode        Unemployed and not actively seeking employment
  Married                                                       446
  Single                                                        175
  Widowed or divorced                                            44
                     Employment_recode
Marital_recode        Working full-time Working part-time
  Married                          3980              1328
  Single                           1299               492
  Widowed or divorced               544               200
```

9 Chi-squared test

When we have a cross-tab with multiple elements, the chi-squared test can check whether the frequency count by cells is likely to be either the result of random scatter, or statistically significant. This indicates some pattern or non-random relationship in the data without explaining it. Like other statistical tools in R, there is a rich set of options and alternatives to work with statistical tests. We are not even scratching the surface here, but including this as a reminder of these capabilities.

```r
chisq.test(table(Direction,Sex))
```

```
    Pearson's Chi-squared test

data:  table(Direction, Sex)
X-squared = 70.858, df = 4, p-value = 1.496e-14
```

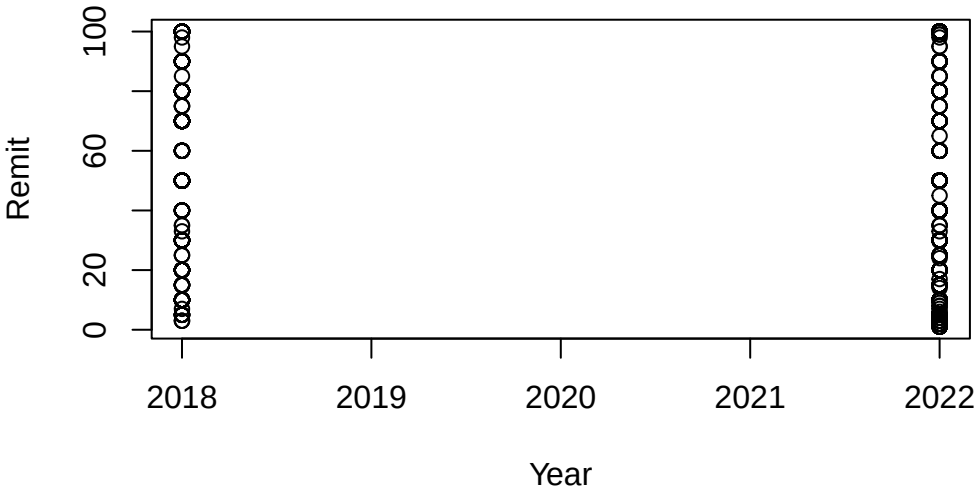
10 Data Visualization

Now it is time to explore the data visually. We explore some variables, in particular focusing on the direction the country is going in. Actually in this case, it is countries, since we are including all three countries, over all three years.

10.1 Scatterplots

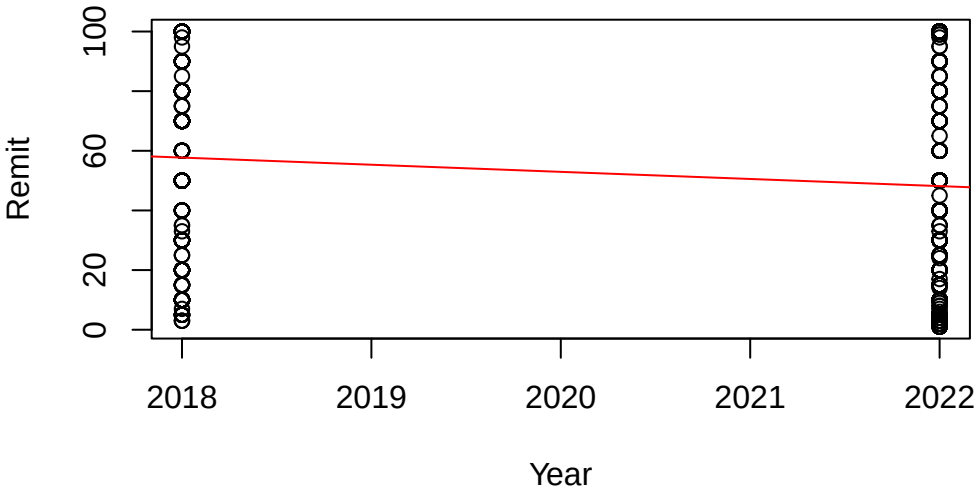
We'll start with scatterplots. In our data, Remit is at least one variable good for scatterplots. Most are not since they are categorical.

```
plot(Remit~Year)
```



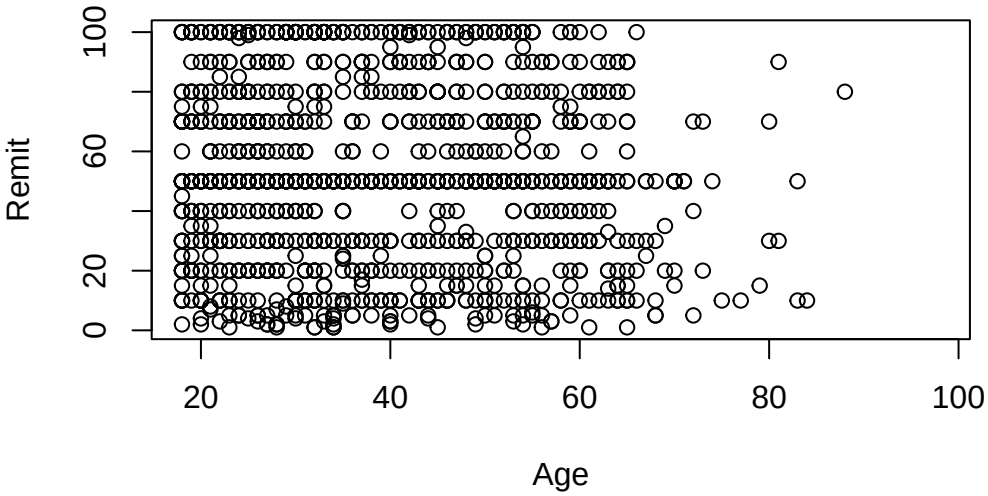
Not very interesting is it? Maybe add a regression line? Still not so interesting.

```
plot(Remit~Year)
abline(lm(Remit~Year), col="red")
```



At least let's choose something to plot against that has some variety.

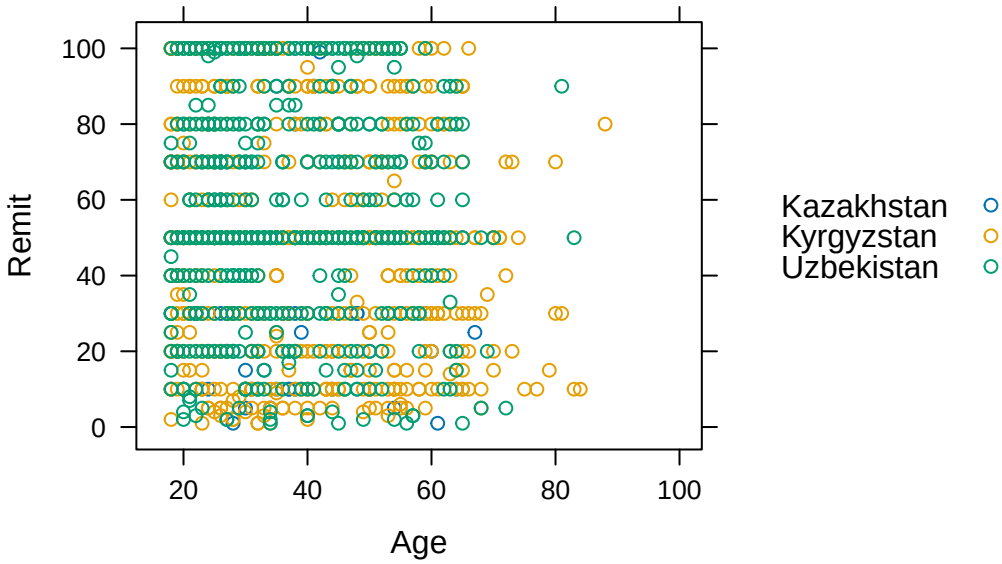
```
plot(Remit~Age)
```



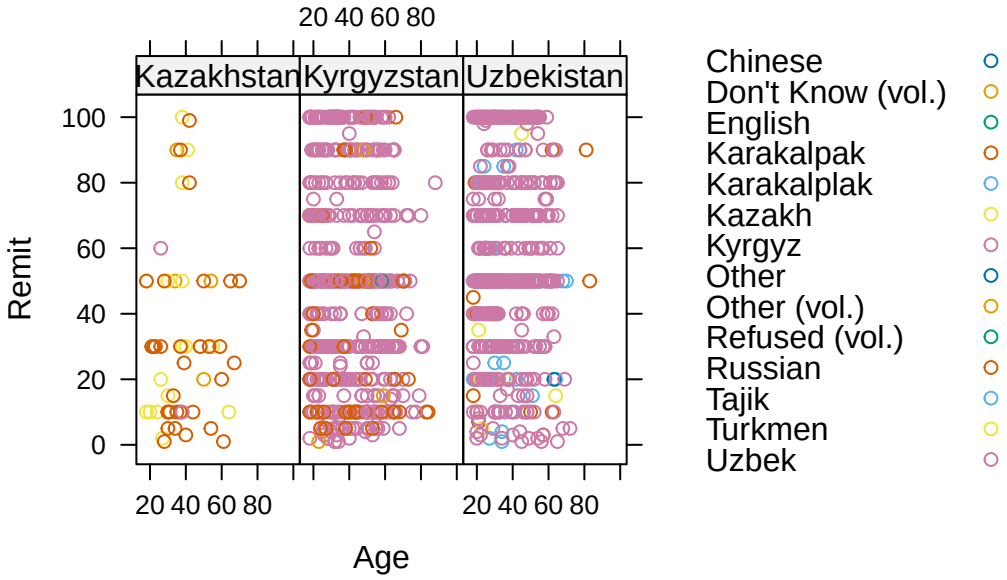
10.2 lattice

Now let's use the *lattice* package to do some more breakdowns of our data. This is one package that has very simple syntax for grouping data.

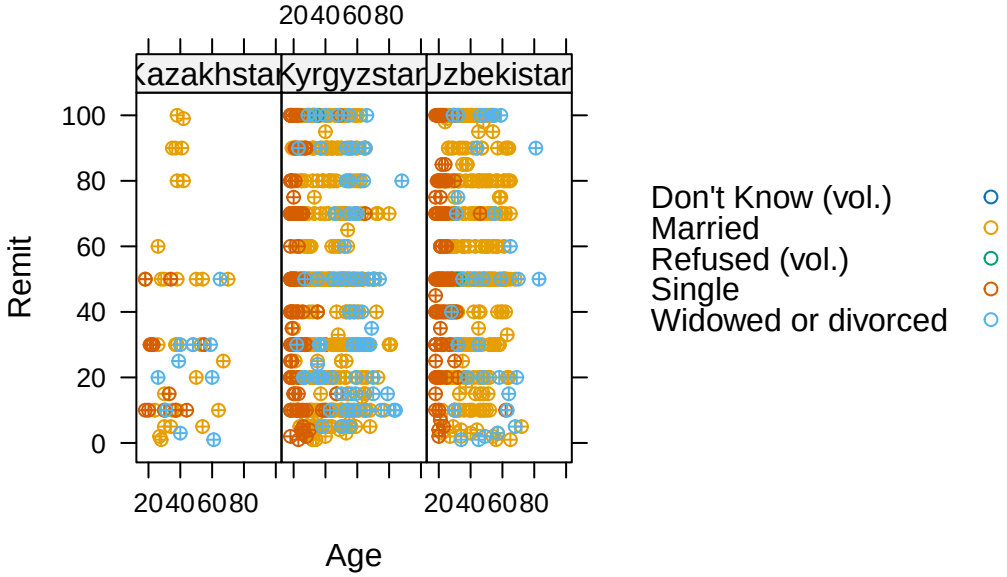
```
xyplot(Remit~Age, groups=Country, auto.key=list(space="right"))
```




```
xyplot(Remit~Age | Country, groups=Language, auto.key=list(space="right"))
```




```
xyplot(Remit~Age | Country, groups=Marital, pch=10,
 ↪  auto.key=list(space="right"))
```



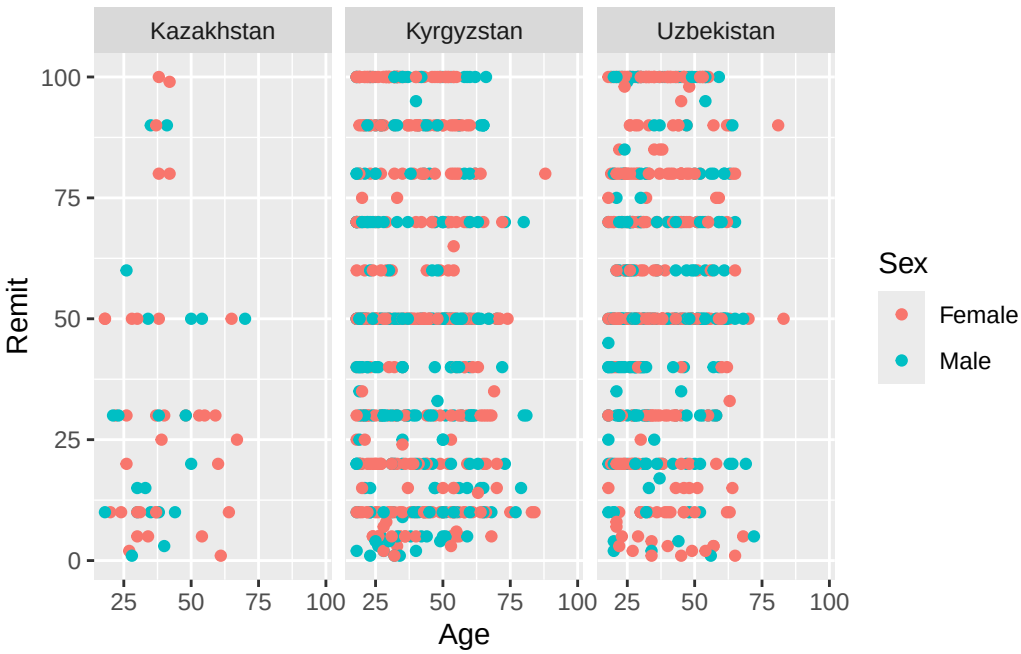
10.3 ggplot

Now let's study the use of *ggplot*, one of the most, if not the most, popular visualization tool in R. It is popular for at least two reasons. First, the default behavior of most graphs is relatively sophisticated and uses good visualization principles. Second, it is very customizable and flexible due to the way the elements of the syntax interact with each other.

First, let's look at the use of "aesthetics", the look and feel element of the "Grammar of Graphics" syntax, and then the use of labels to refine and improve our graphs.

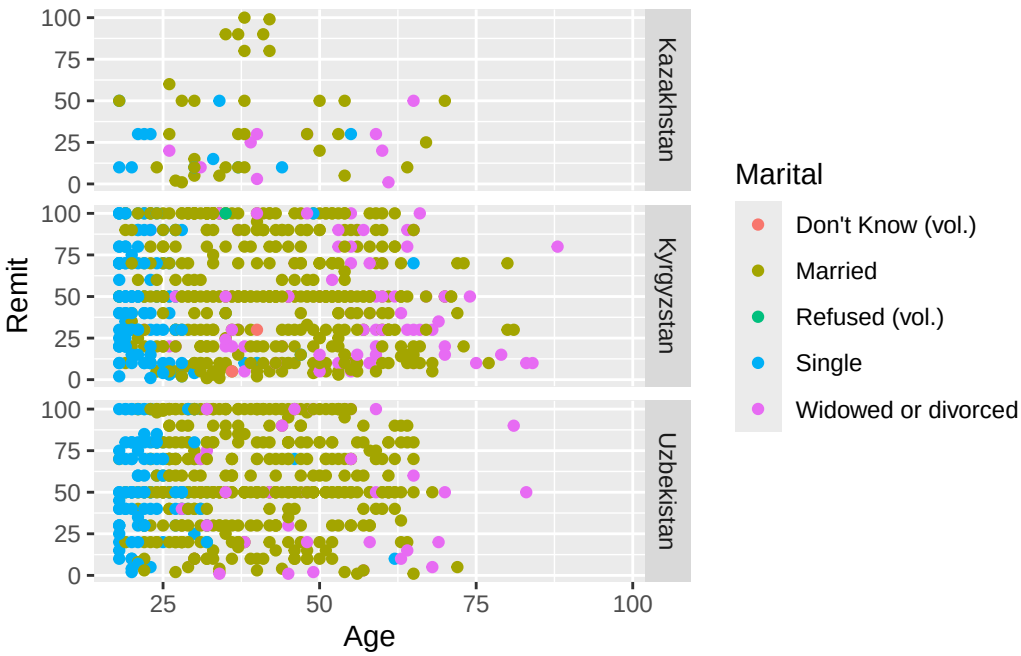
```
ggplot(my_final_data, aes(y=Remit, x=Age))  + facet_grid(.~Country) +
 ↪  geom_point(aes(color=Sex))
```

```
Warning: Removed 13837 rows containing missing values or values outside the scale range
(`geom_point()`).
```



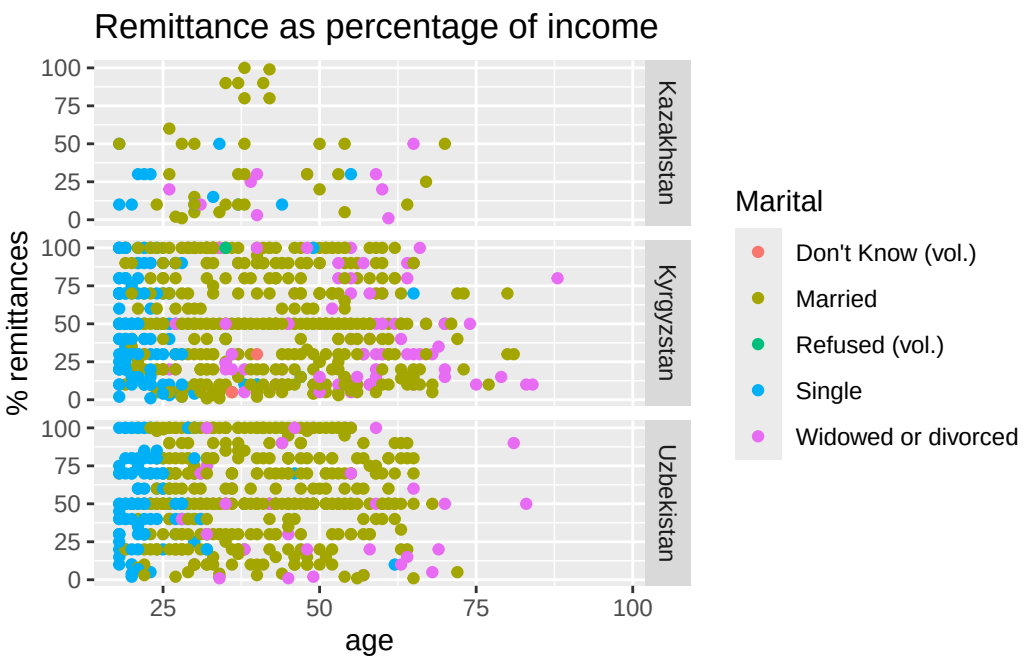

```
ggplot(my_final_data, aes(y=Remit, x=Age))  + facet_grid(Country~.) +
↪  geom_point(aes(color=Marital))
```

```
Warning: Removed 13837 rows containing missing values or values outside the scale range
(`geom_point()`).
```



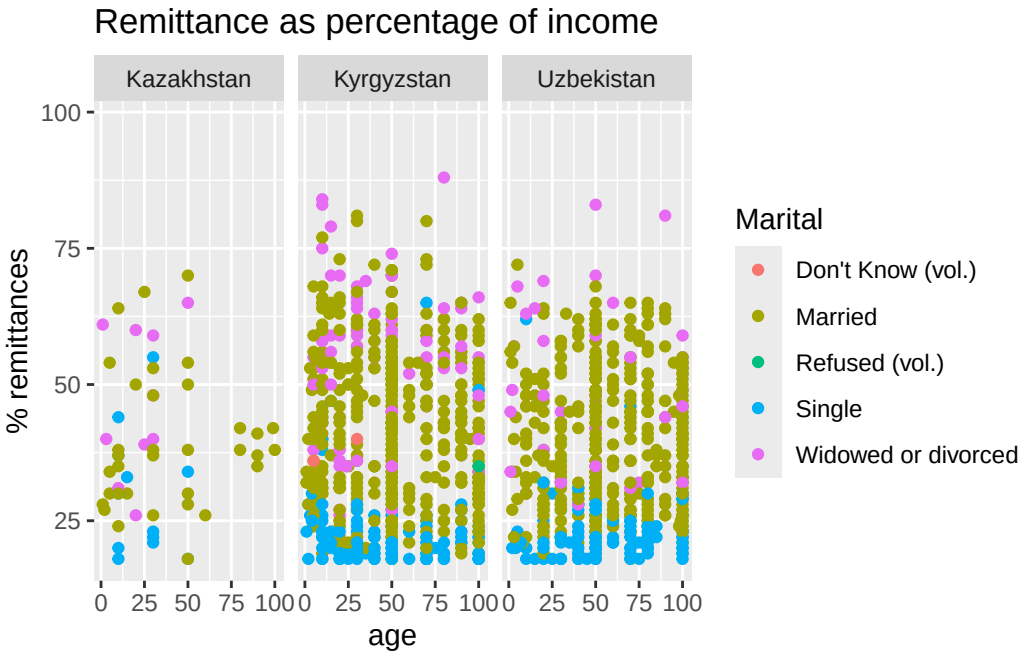
```
ggplot(my_final_data, aes(y=Remit, x=Age))  + facet_grid(Country~.) +
 ↪  geom_point(aes(color=Marital))+
labs(x="age", y="% remittances", title="Remittance as percentage of income")
```

```
Warning: Removed 13837 rows containing missing values or values outside the scale range
(`geom_point()`).
```



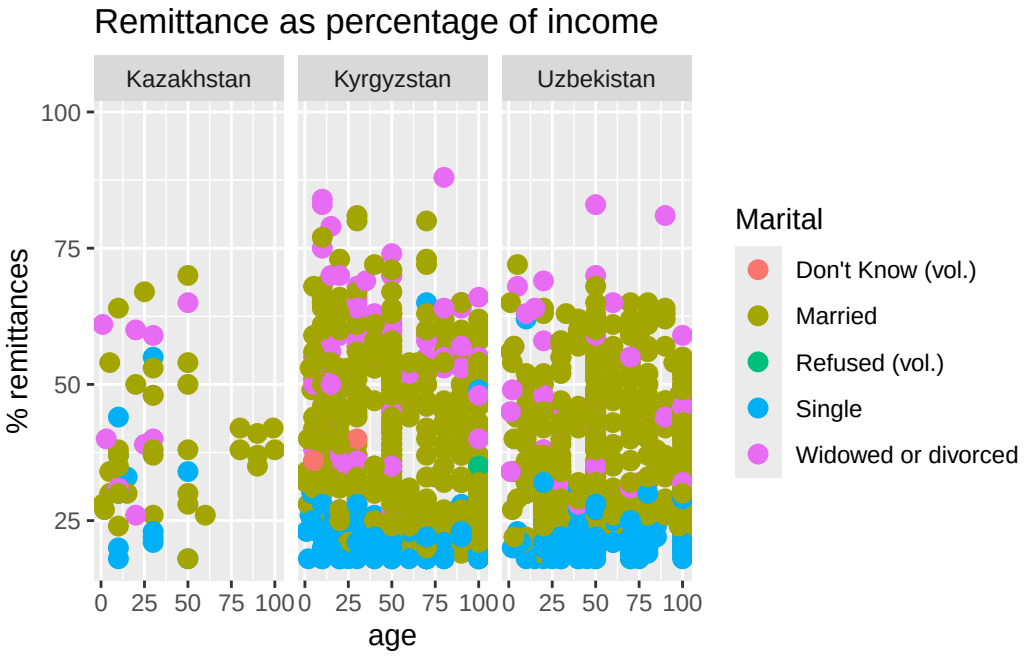

```
ggplot(my_final_data, aes(x=Remit, y=Age))  + facet_grid(.~Country) +
 ↪  geom_point(aes(color=Marital))+
  labs(x="age", y="% remittances", title="Remittance as percentage of
   ↪  income")
```

```
Warning: Removed 13837 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
# point size

ggplot(my_final_data, aes(x=Remit, y=Age))  + facet_grid(.~Country) +
 ↪  geom_point(aes(color=Marital), size=3)+
  labs(x="age", y="% remittances", title="Remittance as percentage of
   ↪  income")
```

```
Warning: Removed 13837 rows containing missing values or values outside the scale range
(`geom_point()`).
```



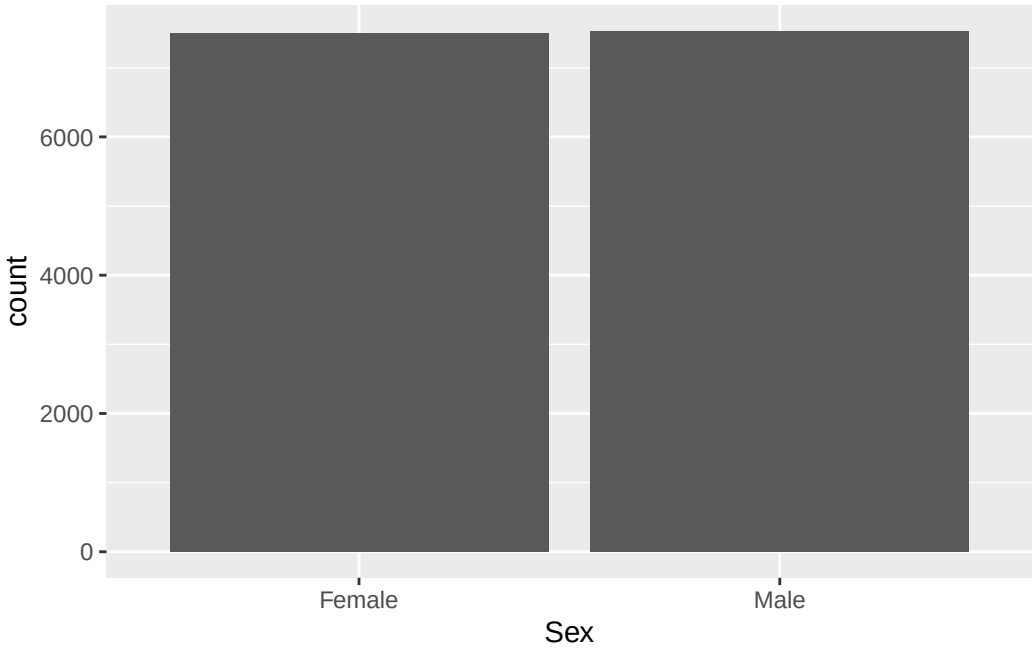
10.4 Bar charts

Note the last tweak there to change the point size, part of the geom_point options. But, since most of our data is categorical, we will get a clearer picture of our data working with bar charts, working our way from very basic, dull examples into examples where we use the power of ggplot to modify the look and feel to convey more complex information and compare our countries and time periods.

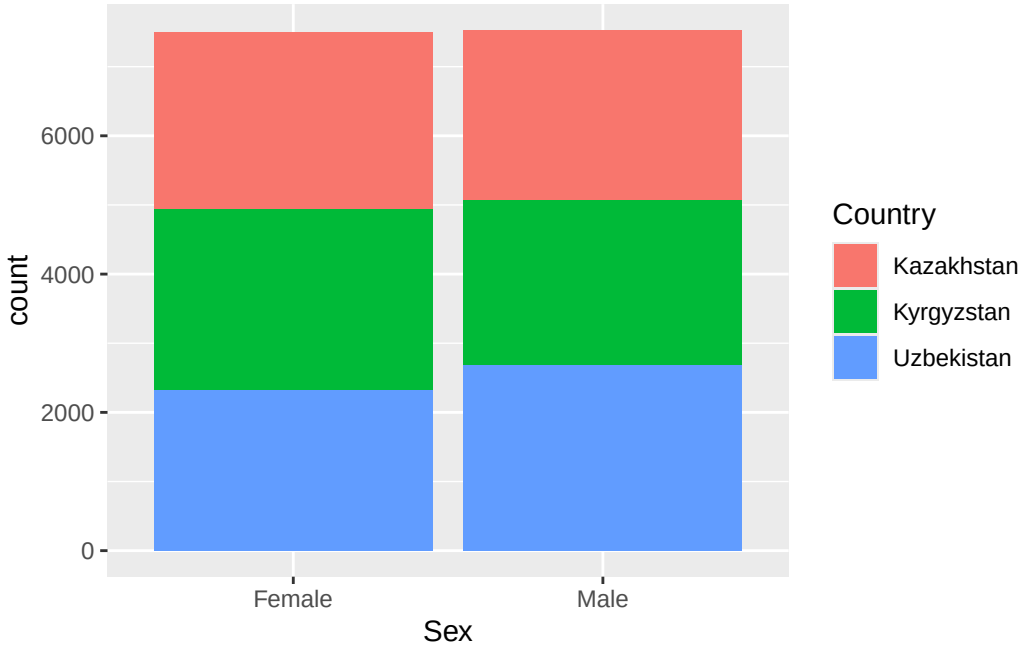
10.5 Colors

The *RColorBrewer* package enables us to apply nicely scaled and shaded sets of colors easily. We can use the R Color Picker to find colors that suit us if we want to use *scale_fill_manual* to provide a color list (in this case matching country flags). Finally, the page dealing with colors in ggplots gives a wealth of information.

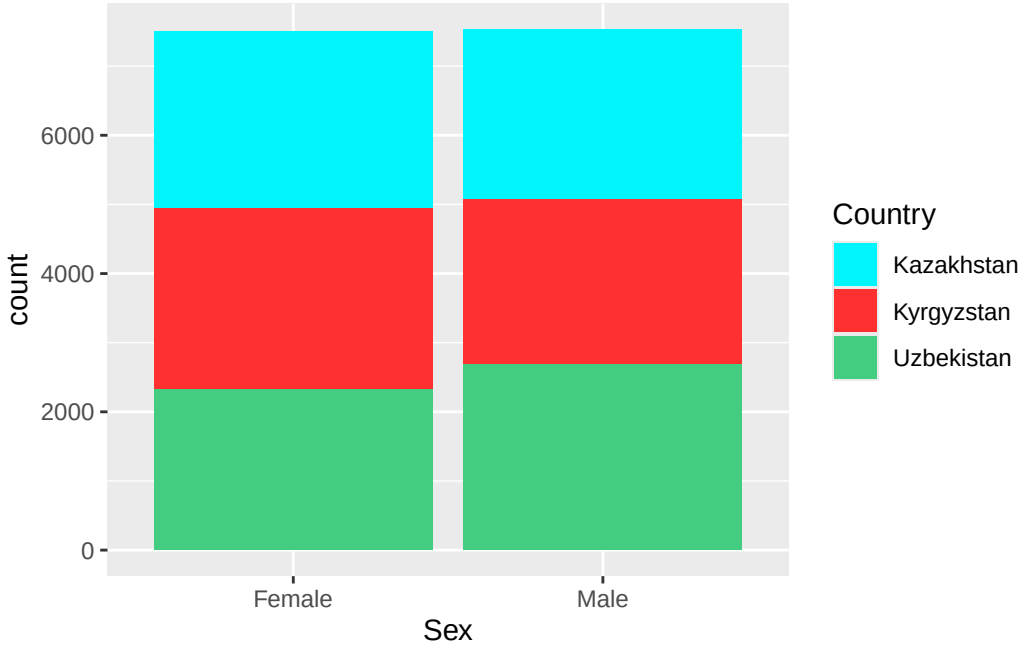
```
ggplot(my_final_data, aes(x=Sex)) + geom_bar(position="stack")
```



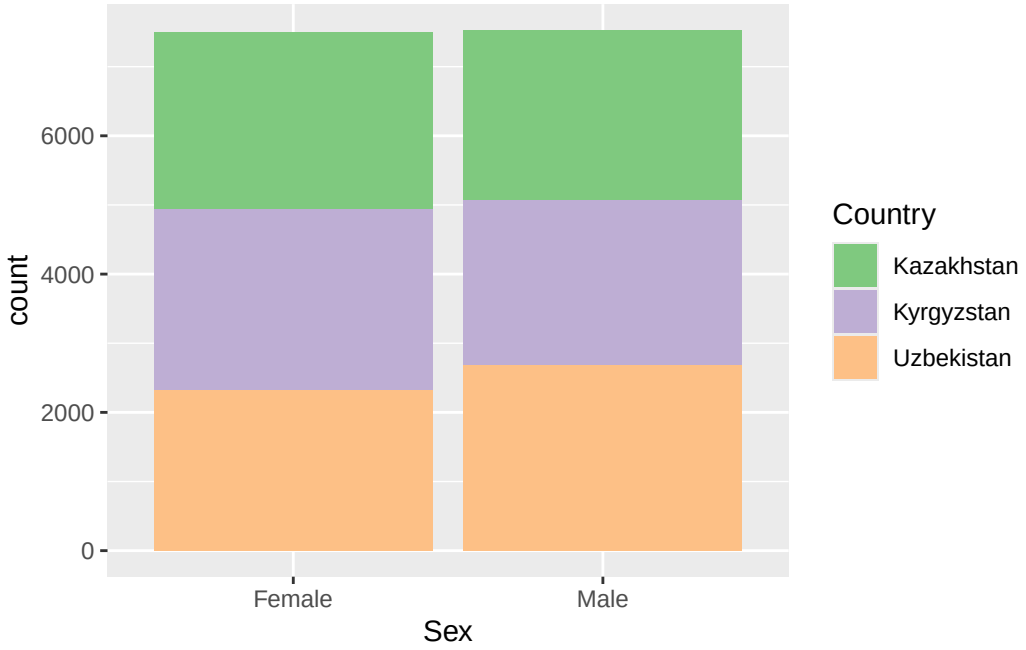

```
ggplot(my_final_data, aes(x=Sex, fill=Country)) + geom_bar(position="stack")
```




```
country_colors <- c("turquoise1","firebrick1","seagreen3")

ggplot(my_final_data, aes(x=Sex, fill=Country)) + geom_bar(position="stack")
 ↪  + scale_fill_manual(values = country_colors)
```

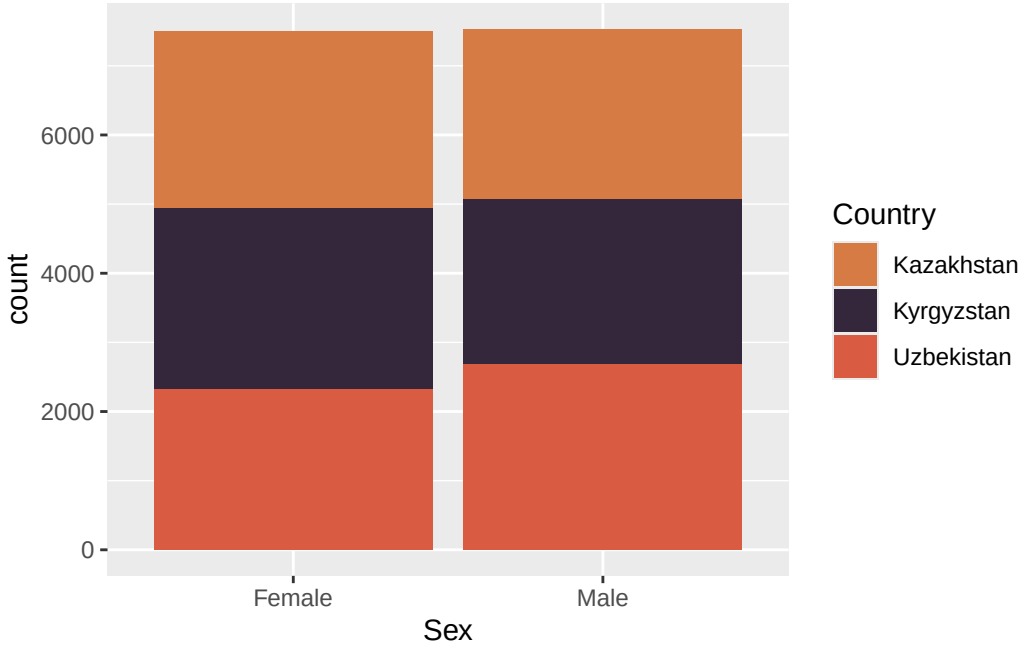



```
ggplot(my_final_data, aes(x=Sex, fill=Country)) + geom_bar(position="stack")
 ↪  + scale_fill_brewer(palette = "Accent")
```

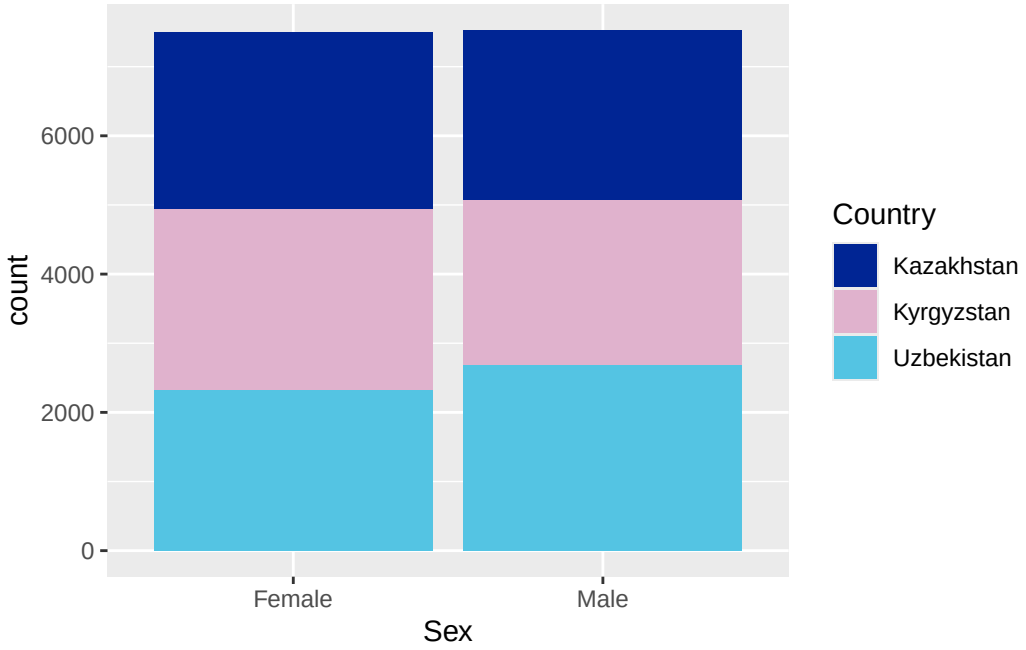


The *paletteer* package introduces a dizzying array of choice into the color selection, but provides tools, including the Color Palette Finder to make it manageable. The palette names sometimes align with the colors they provide. Do you think the *samarqand2* palette matches the colors of the tiles in the Registan?

```
ggplot(my_final_data, aes(x=Sex, fill=Country)) + geom_bar(position="stack")
 ↪  +scale_fill_paletteer_d("nationalparkcolors::Hawaii")
```



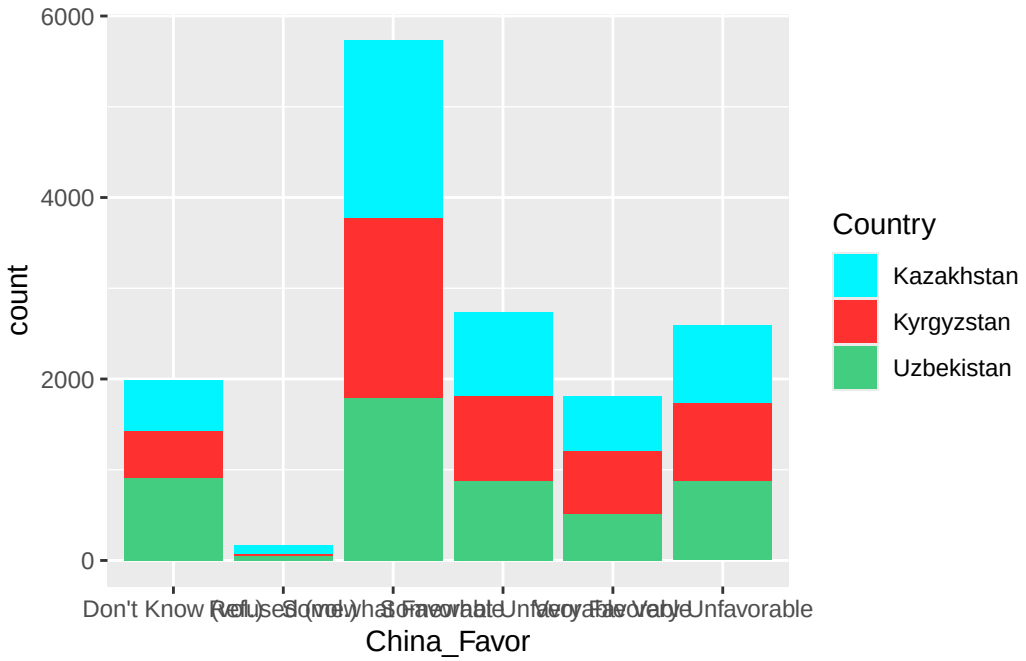

```
ggplot(my_final_data, aes(x=Sex, fill=Country)) + geom_bar(position="stack")
 ↪  +scale_fill_paletteer_d("IslamicArt::samarqand2")
```



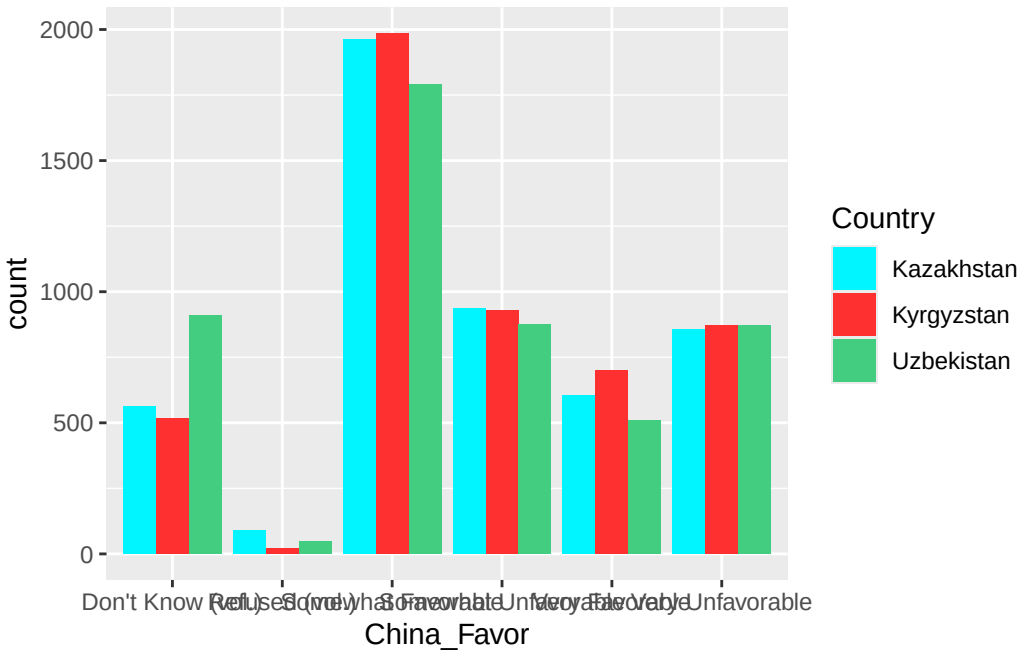

```
# try also
# IslamicArt::samarqand
# nbapalettes::lakers
# MoMAColors::OKeeffe
```

Let's continue the exploration of the data by looking at the Favorability ratings and tweaking the display to our liking.

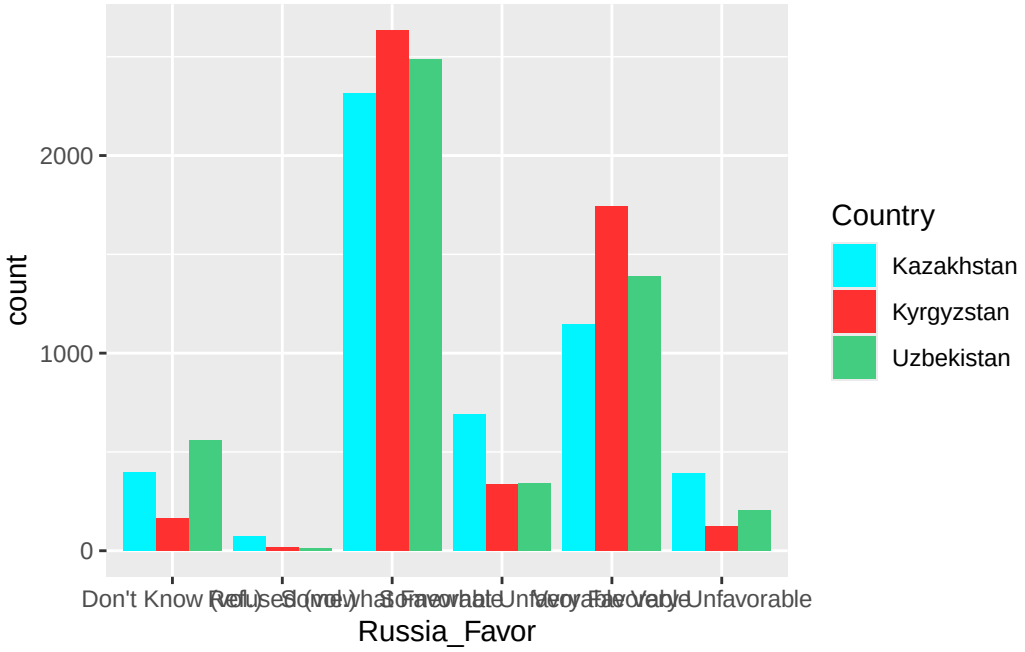
```
ggplot(my_final_data, aes(China_Favor, fill=Country))+
 ↪  geom_bar(position="stack") + scale_fill_manual(values = country_colors)
```



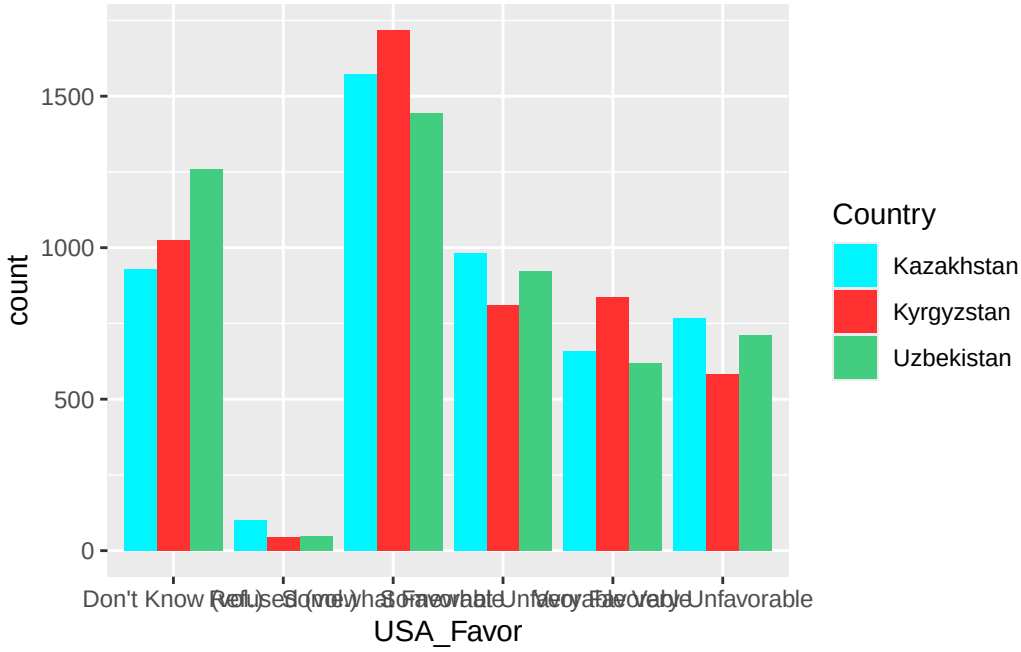
```
ggplot(my_final_data, aes(China_Favor, fill=Country))+
↪  geom_bar(position="dodge") + scale_fill_manual(values = country_colors)
```




```
ggplot(my_final_data, aes(Russia_Favor, fill=Country))+
↪  geom_bar(position="dodge") + scale_fill_manual(values = country_colors)
```




```
ggplot(my_final_data, aes(USA_Favor, fill=Country))+
↪  geom_bar(position="dodge") + scale_fill_manual(values = country_colors)
```

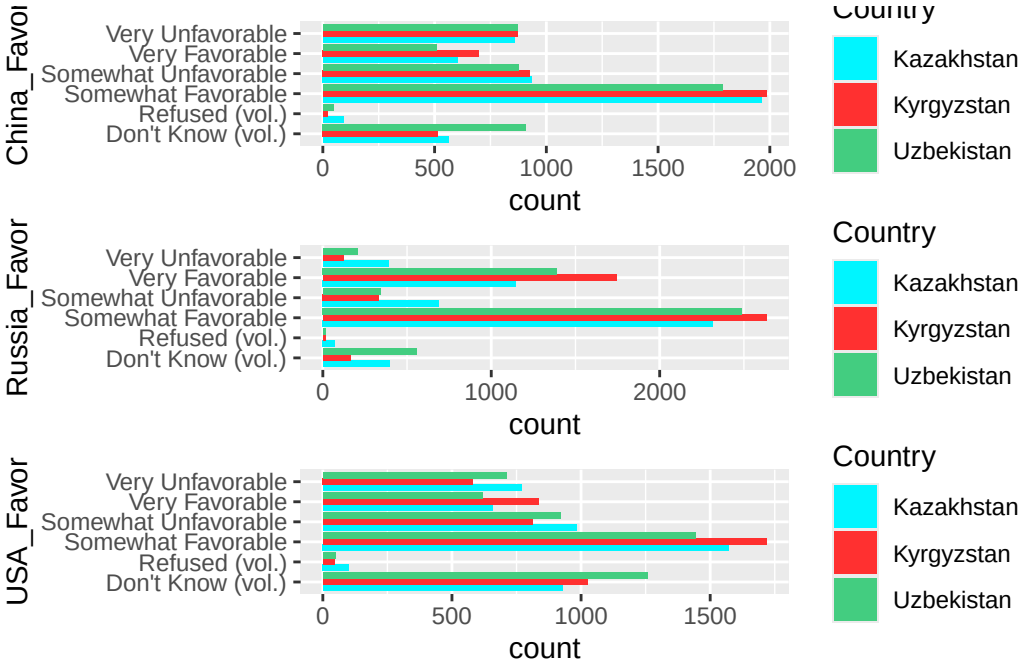


We can export a graph and save it as an object in R using the assignment operator (<-). This will allow us to easily reuse images in different contexts, even to export and modify them.

```
China <- ggplot(my_final_data, aes(China_Favor, fill=Country))+
 ↪  geom_bar(position="dodge") + scale_fill_manual(values = country_colors) +
 ↪  coord_flip()

Russia <- ggplot(my_final_data, aes(Russia_Favor, fill=Country))+
 ↪  geom_bar(position="dodge") + scale_fill_manual(values = country_colors) +
 ↪  coord_flip()

USA <- ggplot(my_final_data, aes(USA_Favor, fill=Country))+
 ↪  geom_bar(position="dodge") + scale_fill_manual(values = country_colors) +
 ↪  coord_flip()
```

Now we'll use the *gridExtra* package and its command *grid.arrange* to combine the three favorability plots into one graphic.

```
grid.arrange(China,Russia,USA, ncol=1, nrow=3)
```



We can export graphs to static saved files on our computer. We initiate the process with a command like *pdf* to send the output to pdf, or *jpeg* to send it to an .jpg file. Then run a graphics command (actually a PDF can hold multiple graphs on additional pages). These graphs do not appear on screen because they are being sent to the file. Then run *dev.off* to stop redirecting output to a file. Afterwards, graphs will appear on screen again.

```
pdf(file="output.pdf")
ggplot(my_final_data, aes(USA_Favor, fill=Country))+
 ↪  geom_bar(position="dodge") + scale_fill_manual(values = country_colors) +
 ↪  coord_flip()
dev.off()
```

```
pdf
  2
```

```
jpeg(file="output.jpg", width = 800, height = 600, quality=100)
ggplot(my_final_data, aes(x=Sex, fill=Country)) + geom_bar(position="stack")
 ↪  +scale_fill_paletteer_d("IslamicArt::samarqand2")
dev.off()
```

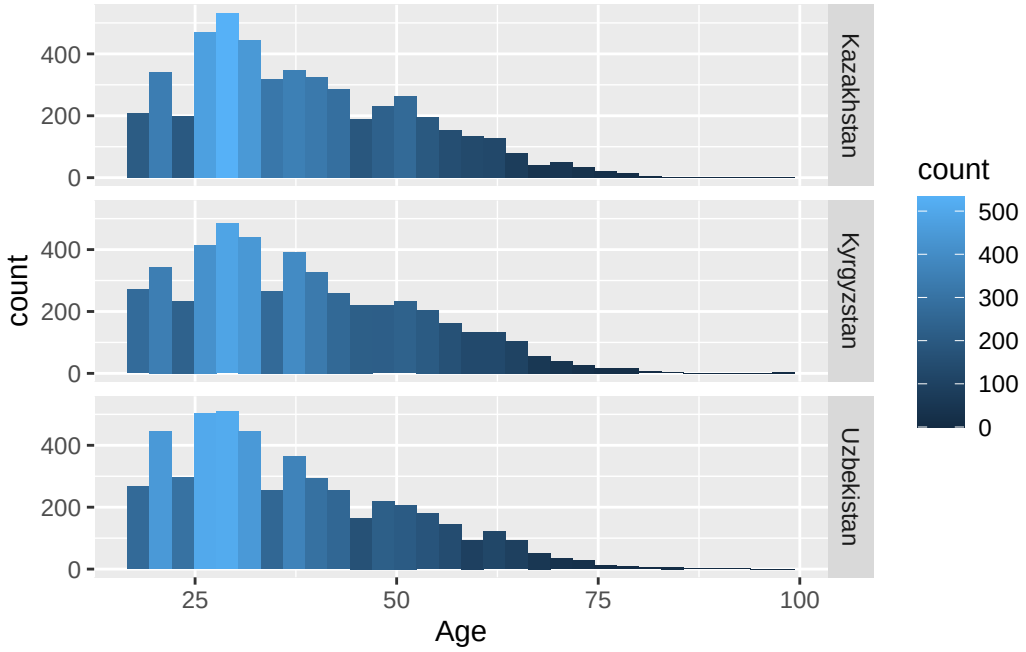
```
pdf
  2
```

10.6 Histogram

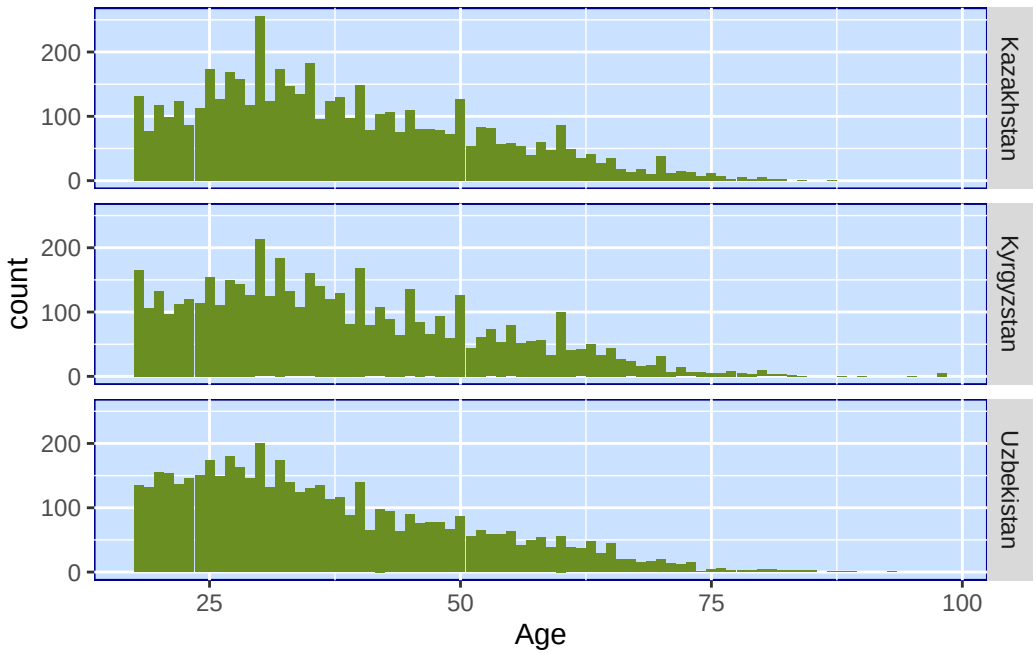
To show some other possibilities, let's look at histograms and how to tweak them. Here we also illustrate the use of the *theme* option.

```
ggplot(my_final_data, aes(Age))+geom_histogram(aes(fill =
↪  after_stat(count)))+facet_grid(Country~.)
```

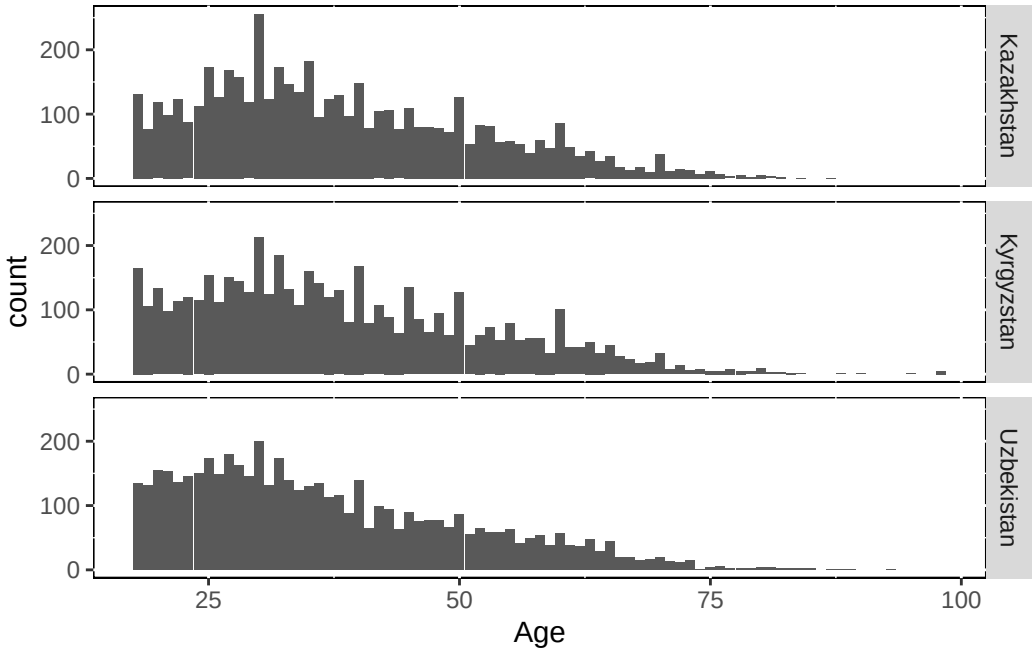
```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



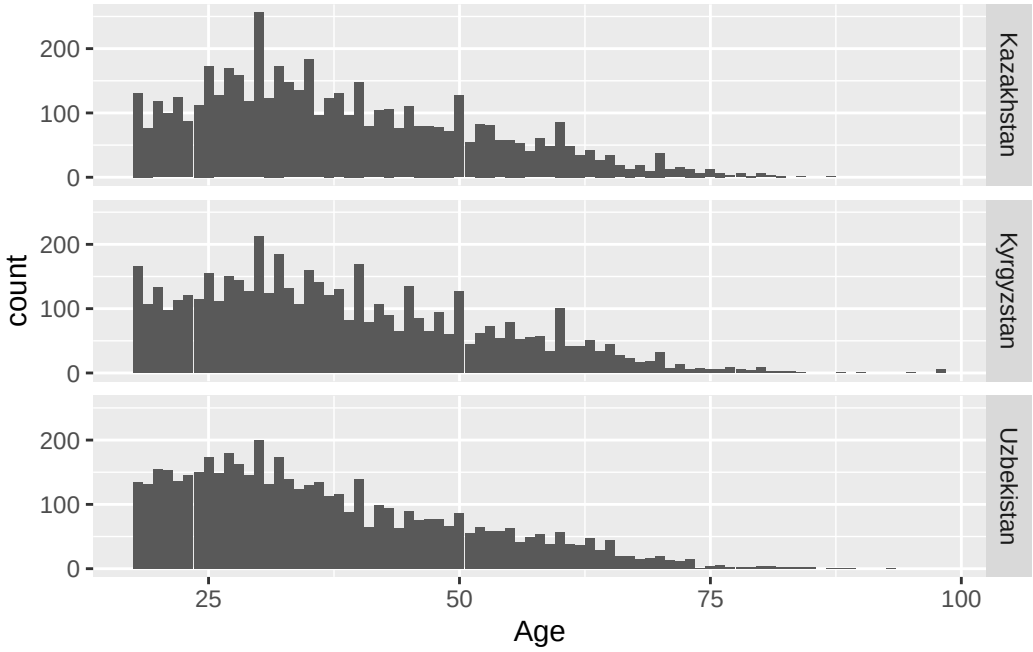

```
ggplot(my_final_data, aes(Age)) + facet_grid(Country~.) +
↪  geom_bar(position="dodge", fill="olivedrab")+theme(panel.background =
↪  element_rect(fill='lightsteelblue1', colour='darkblue'))
```



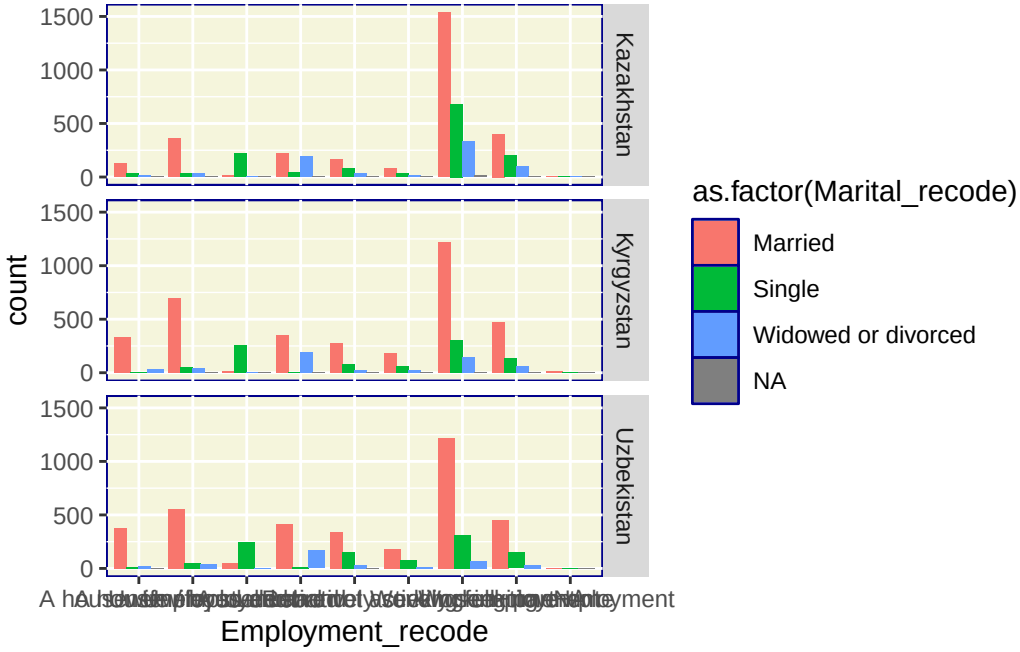
```
ggplot(my_final_data, aes(Age)) + facet_grid(Country~.) +
↪  geom_bar(position="dodge")+theme(panel.background =
↪  element_rect(fill='white', colour='black'))
```



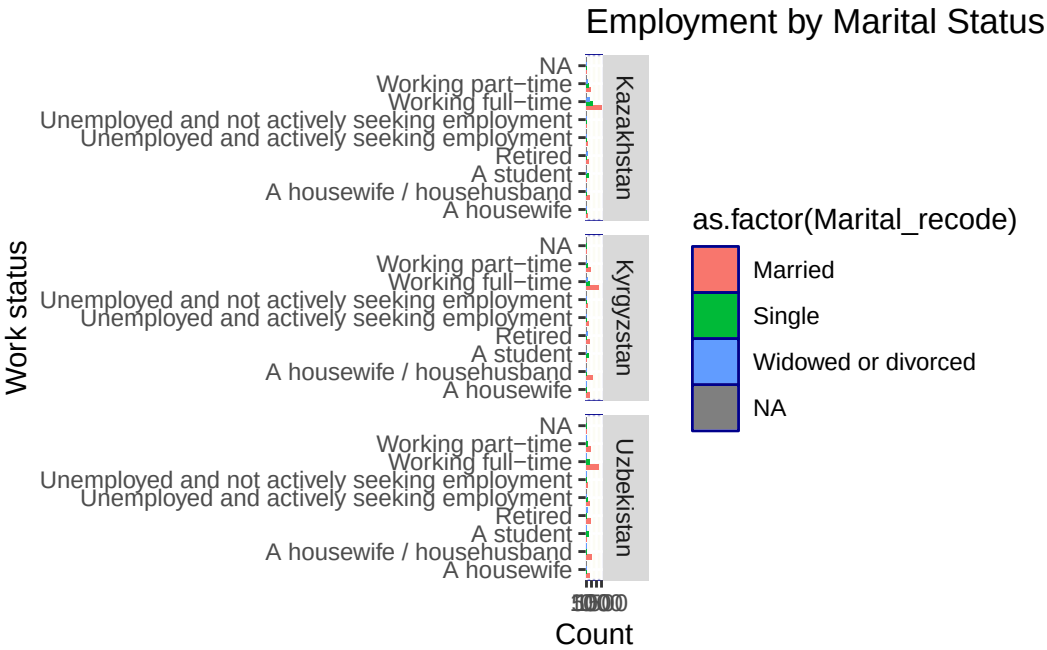

```
ggplot(my_final_data, aes(Age)) + facet_grid(Country~.) +
↪  geom_bar(position="dodge")+ scale_fill_brewer(palette="Reds")
```




```
ggplot(my_final_data, aes(Employment_recode, fill=as.factor(Marital_recode)))
↪  +facet_grid(Country~.) + geom_bar(position="dodge") +
↪  theme(panel.background = element_rect(fill='beige', colour='darkblue'))
```

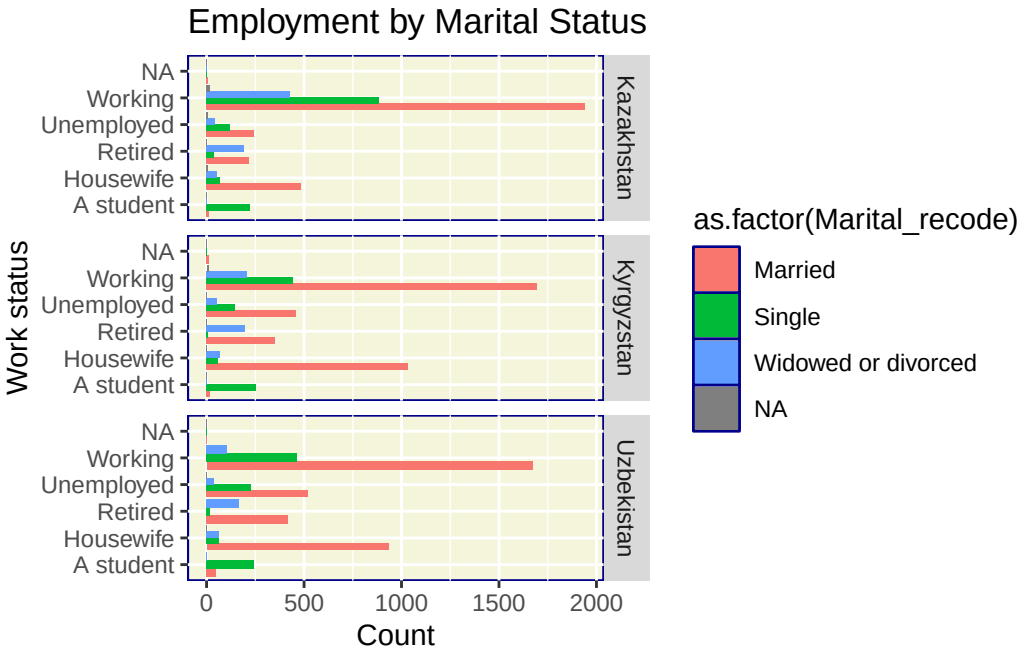



```
ggplot(my_final_data, aes(Employment_recode, fill=as.factor(Marital_recode)))
↪  +facet_grid(Country~.) + geom_bar(position="dodge") +
↪  theme(panel.background = element_rect(fill='beige', colour='darkblue')) +
 labs(x="Work status", y="Count", title="Employment by Marital Status") +
  ↪  coord_flip()
```

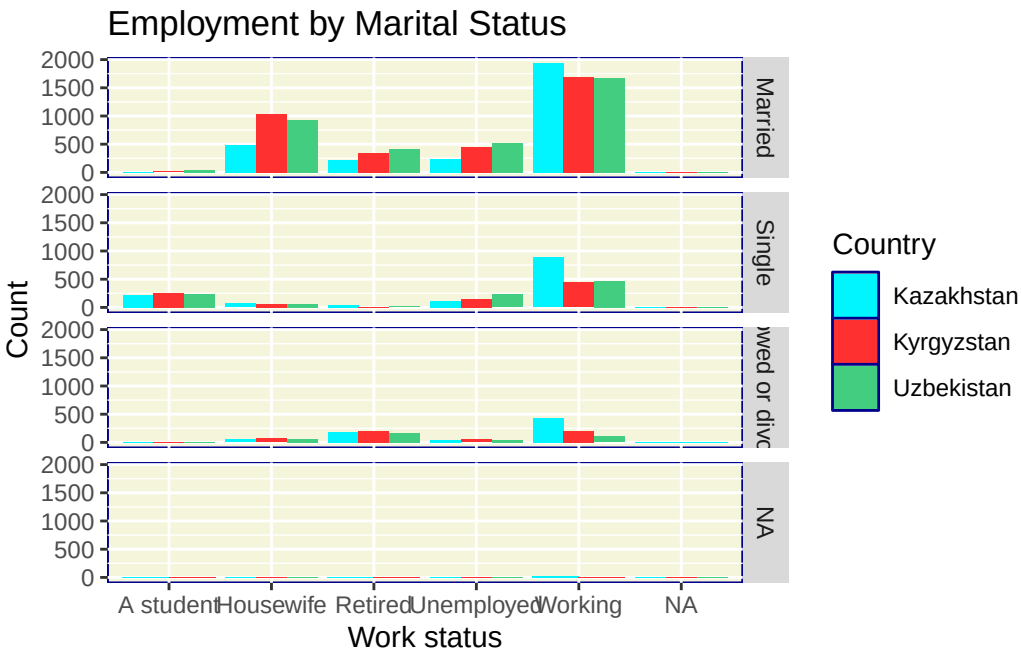



```
ggplot(my_final_data, aes(Employment_recode2,
↪  fill=as.factor(Marital_recode))) +facet_grid(Country~.) +
↪  geom_bar(position="dodge") + theme(panel.background =
↪  element_rect(fill='beige', colour='darkblue')) +
```

```
  labs(x="Work status", y="Count", title="Employment by Marital Status") +
 ↪  coord_flip()
```



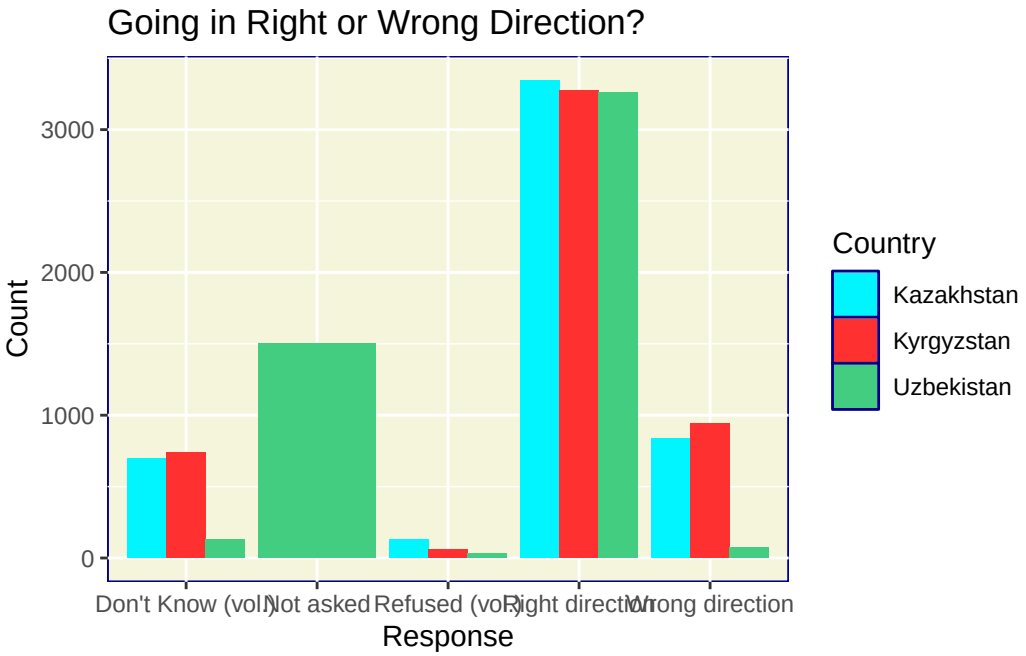

```
ggplot(my_final_data, aes(Employment_recode2, fill=Country))
 ↪  +facet_grid(Marital_recode~.) + geom_bar(position="dodge") +
 ↪  scale_fill_manual(values=country_colors)+theme(panel.background =
 ↪  element_rect(fill='beige', colour='darkblue')) +
  labs(x="Work status", y="Count", title="Employment by Marital Status")
```



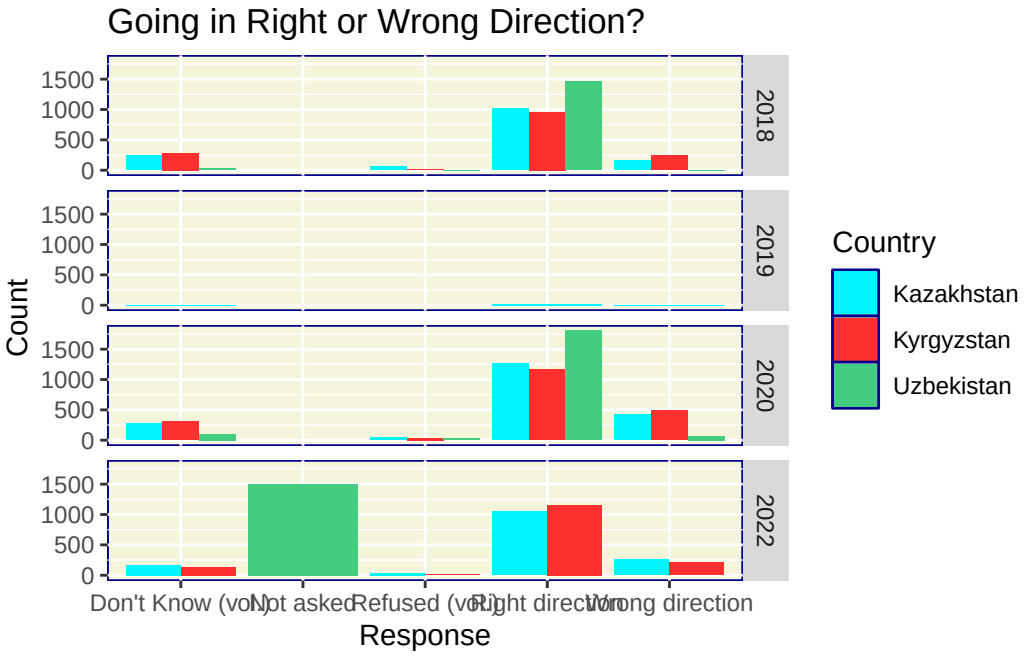
Study the direction question some more:

```
# Direction by factors
```

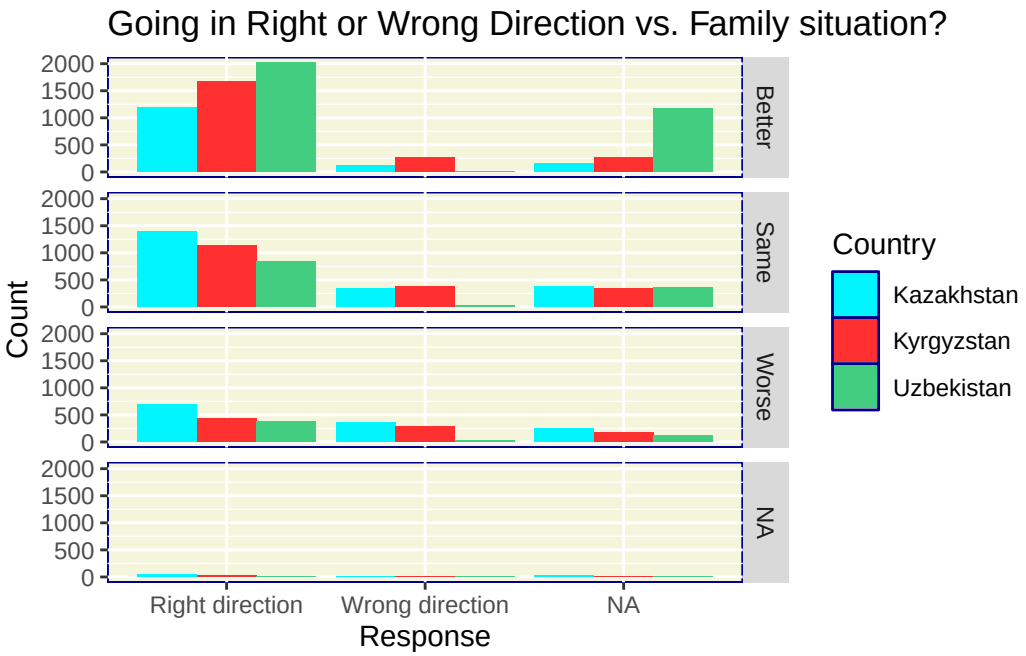
```
ggplot(my_final_data, aes(Direction, fill=Country)) +
↪  geom_bar(position="dodge") +
↪  scale_fill_manual(values=country_colors)+theme(panel.background =
↪  element_rect(fill='beige', colour='darkblue')) +
  labs(x="Response", y="Count", title="Going in Right or Wrong Direction?")
```




```
ggplot(my_final_data, aes(Direction, fill=Country)) + facet_grid(Year~.) +
↪  geom_bar(position="dodge") +
↪  scale_fill_manual(values=country_colors)+theme(panel.background =
↪  element_rect(fill='beige', colour='darkblue')) +
  labs(x="Response", y="Count", title="Going in Right or Wrong Direction?")
```



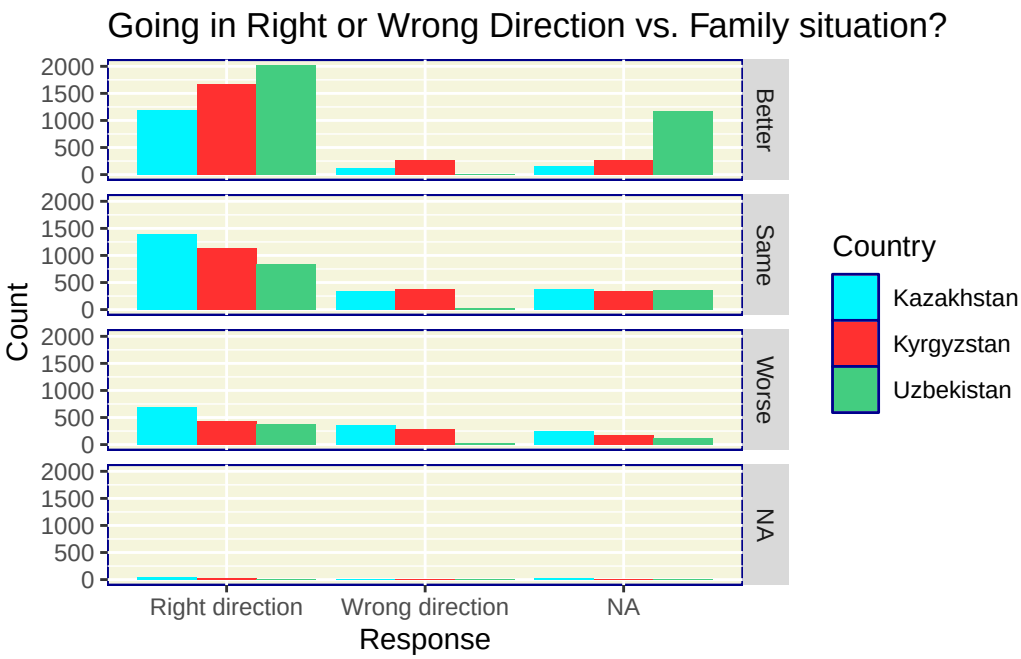

```
ggplot(my_final_data, aes(Direction_recode, fill=Country)) +
↪  facet_grid(Family_Fin_Sit_recode~.) + geom_bar(position="dodge") +
↪  scale_fill_manual(values=country_colors) + theme(panel.background =
↪  element_rect(fill='beige', colour='darkblue')) +
 labs(x="Response", y="Count", title="Going in Right or Wrong Direction vs.
  ↪  Family situation?")
```



10.7 Using the power

We can use the power of the grammar of graphics and assign components of the syntax to different objects, and then string them together. This can help us declutter code and simplify changes and edits (for example, by modifying the theme of all graphs at once).

```
mydata <- ggplot(my_final_data, aes(Direction_recode, fill=Country)) +
 ↪  facet_grid(Family_Fin_Sit_recode~.) +
  labs(x="Response", y="Count", title="Going in Right or Wrong Direction vs.
   ↪  Family situation?") + scale_fill_manual(values=country_colors)

mytheme <- theme(panel.background = element_rect(fill='beige',
 ↪  colour='darkblue'))

mychart <- geom_bar(position="dodge")

mydata + mytheme + mychart
```

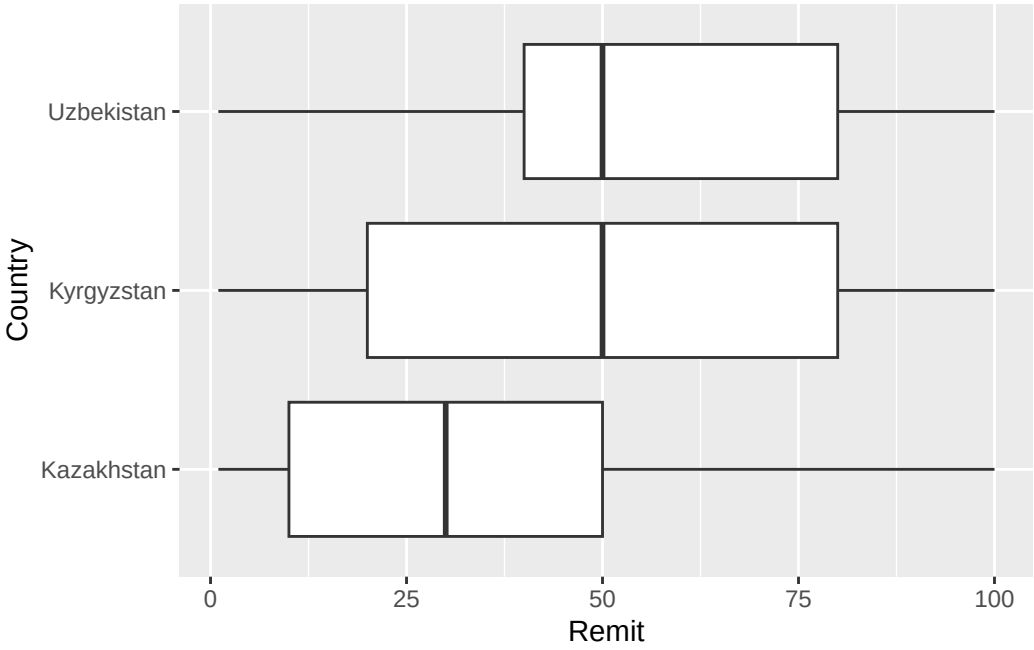


10.8 Other plots

Many other plots are available through ggplot2. A boxplot is shown below. Others can be found at the ggplot2 reference page and the extensions to ggplot2. Note that the *ggextra* package can add boxplots to the margins of scatterplots. Note the themes exist in packages like *ggtheme* and *cowplot*.

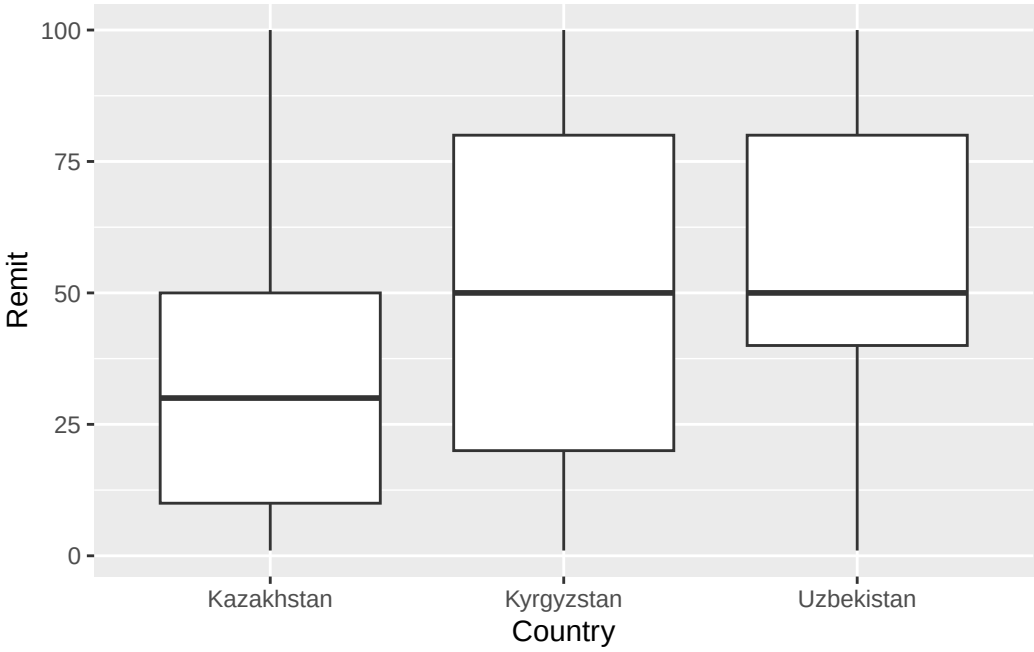
```
ggplot(my_final_data, aes(x=Remit, y=Country)) + geom_boxplot()
```

```
Warning: Removed 13837 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



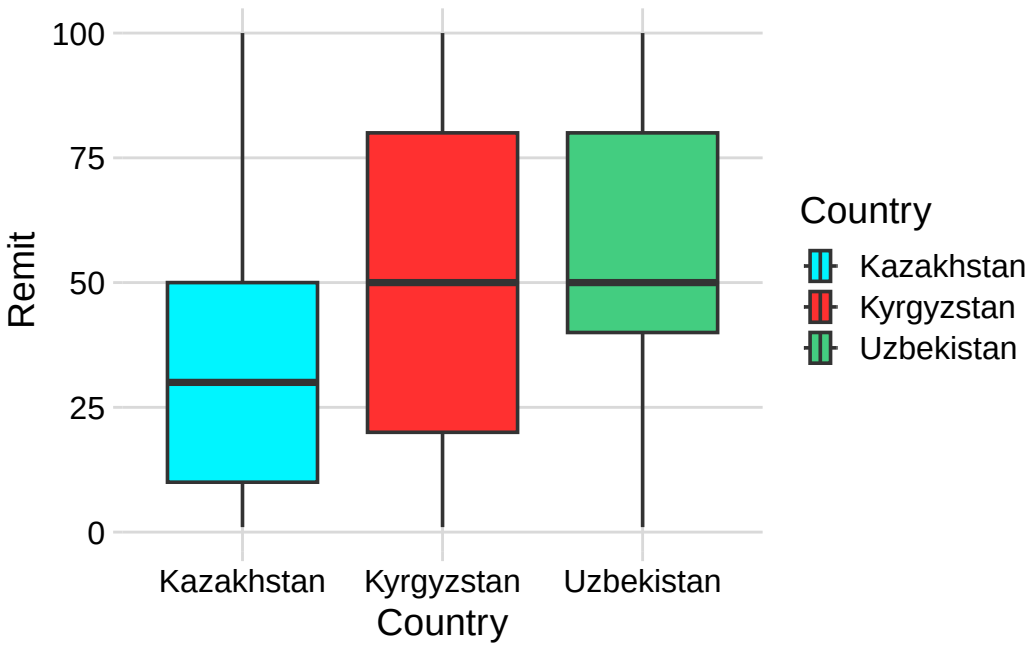

```
ggplot(my_final_data, aes(x=Remit, y=Country)) + geom_boxplot() +
↪  coord_flip()
```

```
Warning: Removed 13837 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```




```
ggplot(my_final_data, aes(x=Remit, y=Country, fill=Country)) + geom_boxplot()
↪  + coord_flip() +
  theme_minimal_grid() + scale_fill_manual(values=country_colors)
```

```
Warning: Removed 13837 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



10.9 Themes

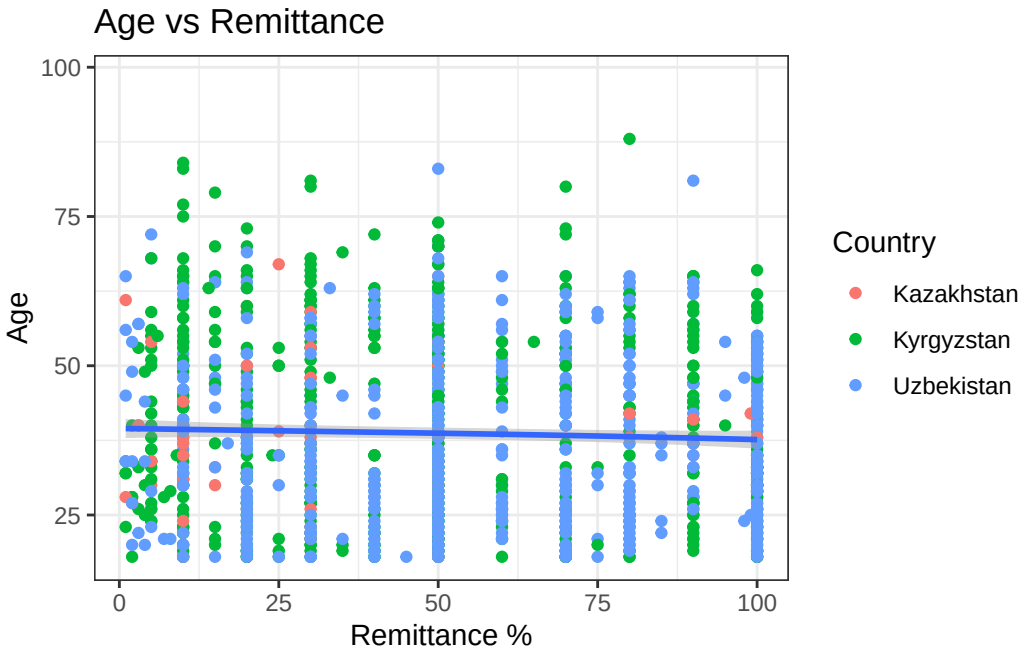
Add-on packages such as ggthemes (illustrated below), ggsci, ggthemr, and more, help us to quickly change the look and feel of our graphs to suit our purpose.

```
lastplot <- ggplot(my_final_data, aes(x=Remit,y=Age))  +
 ↪  geom_point(aes(color=Country)) + stat_smooth() +
  labs(x="Remittance %", y="Age", title="Age vs Remittance")

lastplot + theme_bw()
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

```
Warning: Removed 13837 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

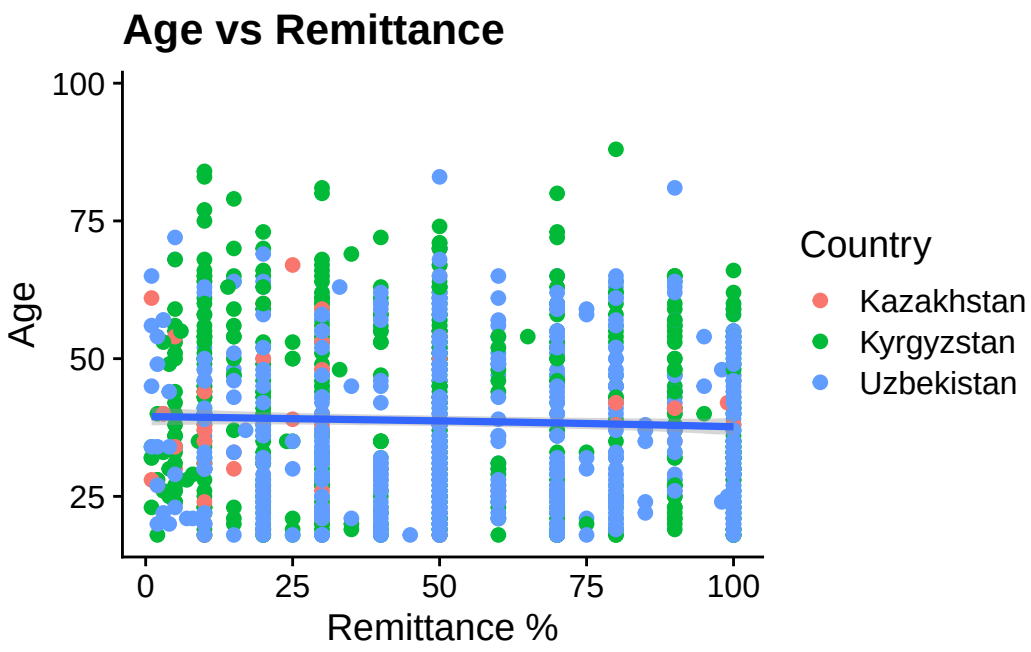
```
Warning: Removed 13837 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
lastplot + theme_cowplot()
```

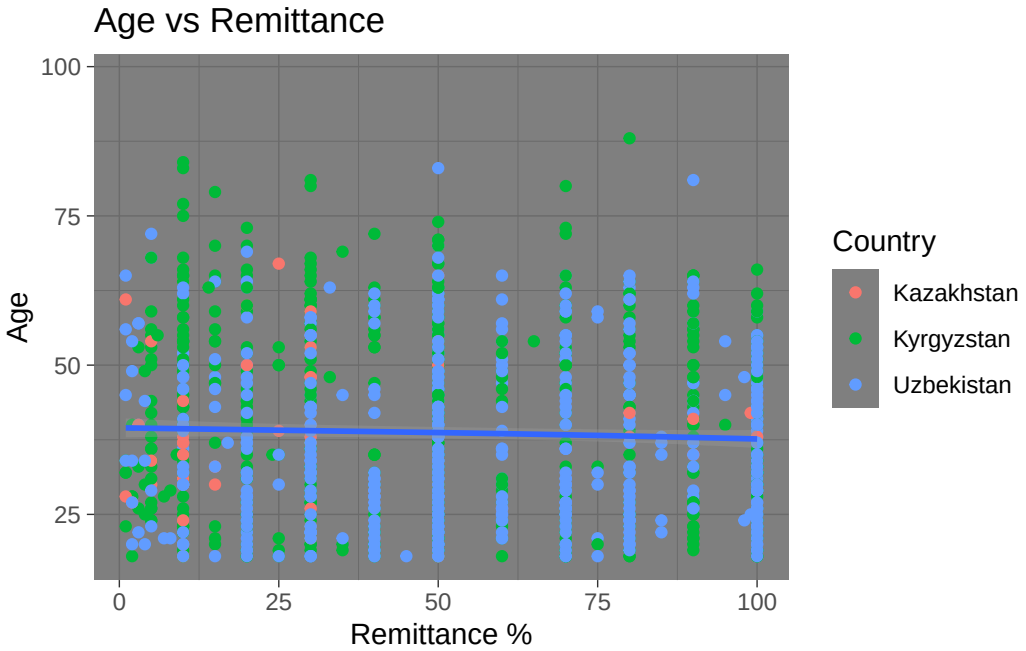
```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

```
Warning: Removed 13837 rows containing non-finite outside the scale range
(`stat_smooth()`).
Removed 13837 rows containing missing values or values outside the scale range
(`geom_point()`).
```

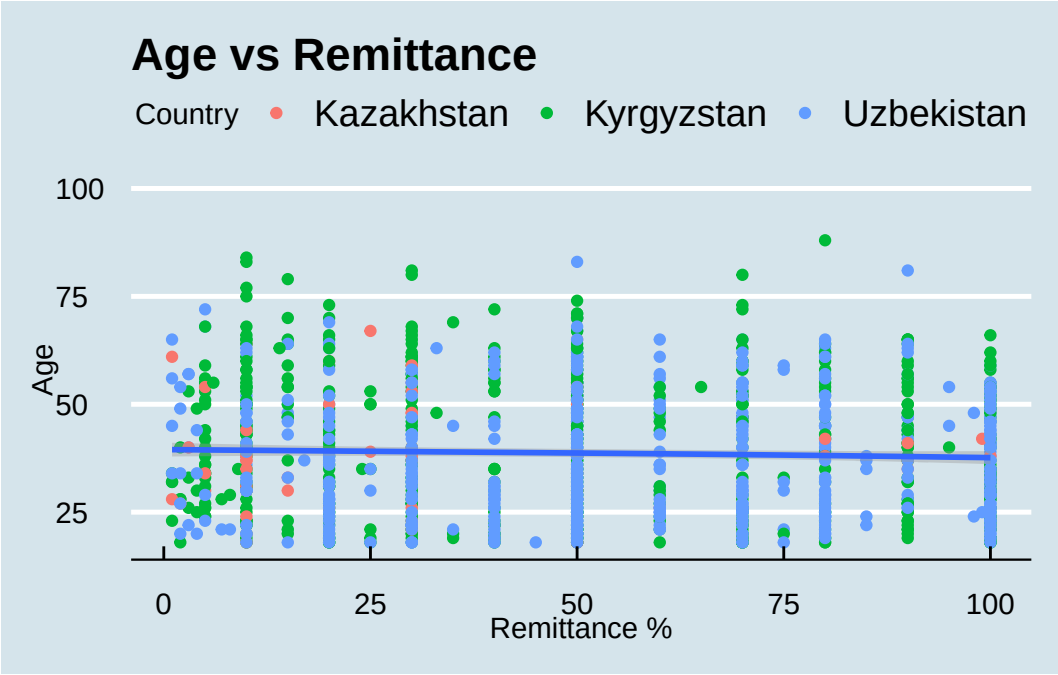



```
lastplot + theme_dark()
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'

Warning: Removed 13837 rows containing non-finite outside the scale range
(`stat_smooth()`).
Removed 13837 rows containing missing values or values outside the scale range
(`geom_point()`).
```



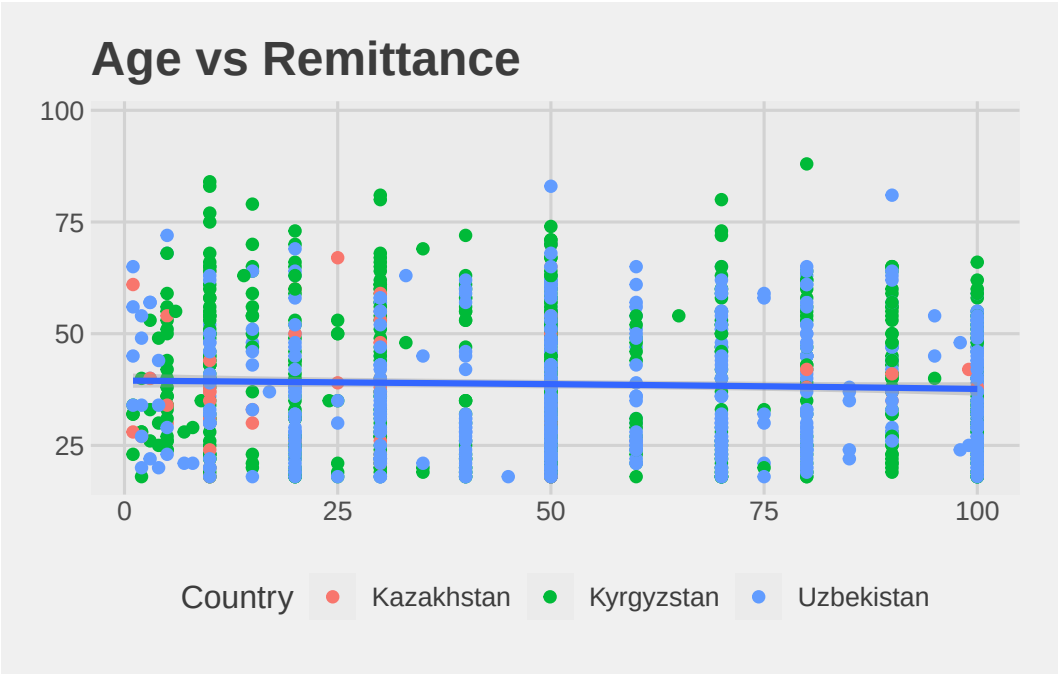

```
lastplot + theme_economist()

`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'

Warning: Removed 13837 rows containing non-finite outside the scale range
(`stat_smooth()`).
Removed 13837 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
lastplot + theme_fivethirtyeight()
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

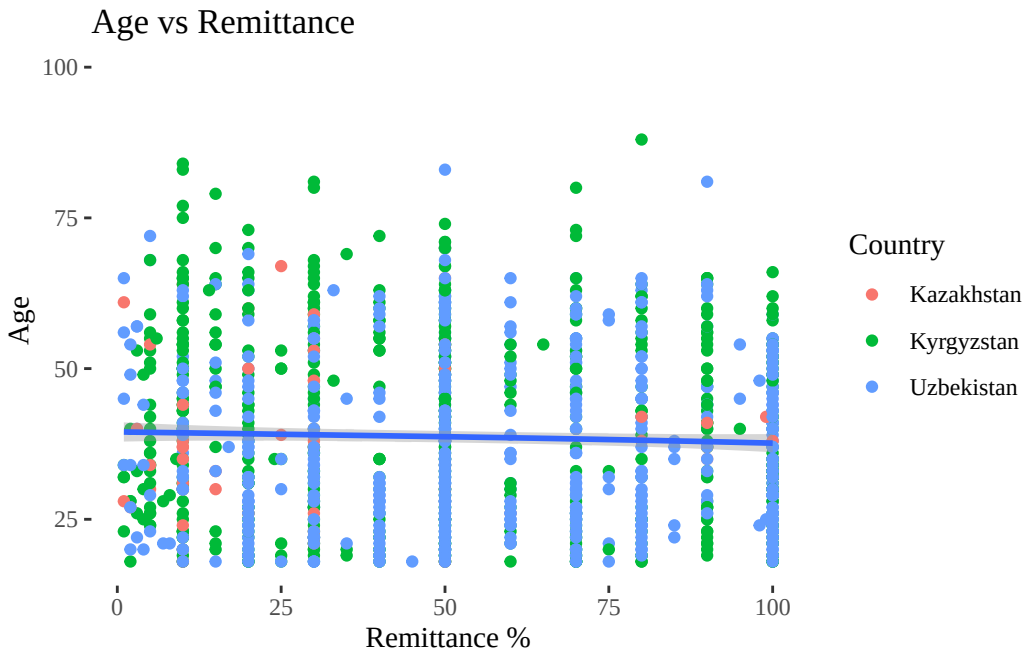
```
Warning: Removed 13837 rows containing non-finite outside the scale range
(`stat_smooth()`).
Removed 13837 rows containing missing values or values outside the scale range
(`geom_point()`).
```



```
lastplot + theme_tufte()
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

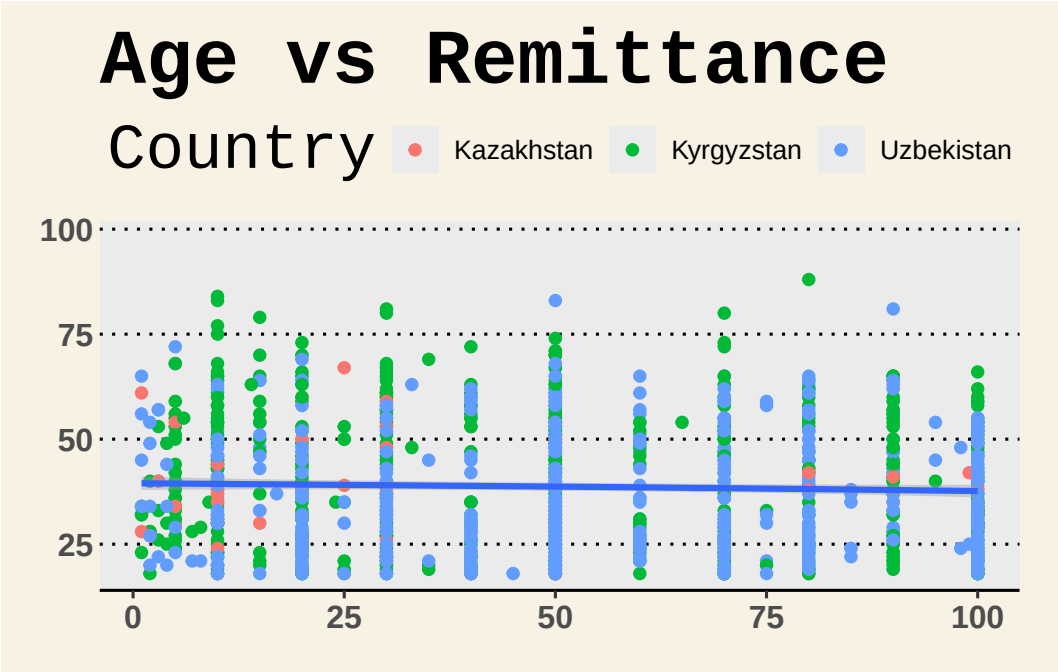
```
Warning: Removed 13837 rows containing non-finite outside the scale range
(`stat_smooth()`).
Removed 13837 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
lastplot + theme_wsj()
```

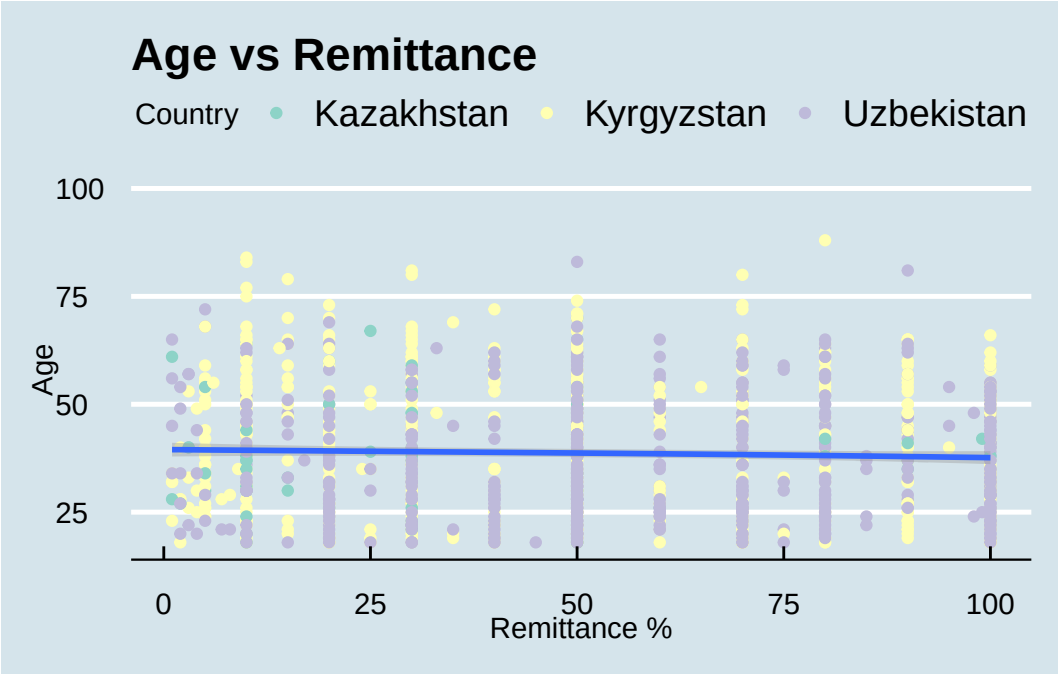
```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

```
Warning: Removed 13837 rows containing non-finite outside the scale range
(`stat_smooth()`).
Removed 13837 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
# then with Rcolorbrewer

lastplot + theme_economist() + scale_color_brewer(palette = "Set3")
```

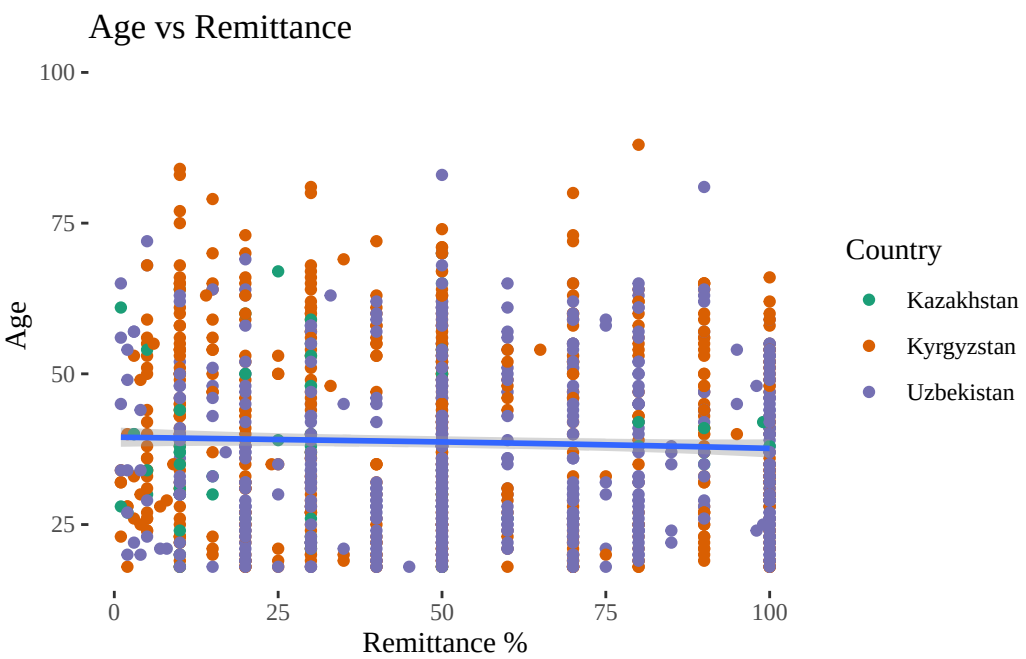
```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'

Warning: Removed 13837 rows containing non-finite outside the scale range
(`stat_smooth()`).
Removed 13837 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
lastplot + theme_tufte() + scale_color_brewer(palette = "Dark2")
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

```
Warning: Removed 13837 rows containing non-finite outside the scale range
(`stat_smooth()`).
Removed 13837 rows containing missing values or values outside the scale range
(`geom_point()`).
```



10.10 Interactivity

We give a quick sketch of introducing a bit of interactivity into a graph. There are tools to build fully interactive sites, particular Shiny for R (and now Python). See my Data Visualization 2 workshop for more.

```
myplot <- ggplot(
  data = my_final_data,
  mapping = aes(
    x = Remit, y = Age,
    # here we add interactive aesthetics
    tooltip = paste(Country,Marital)
  )
) +
  geom_point_interactive(
    size = 1, hover_nearest = TRUE
  )

interactive_plot <- girafe(ggobj = myplot)
```

```
Warning: Removed 13837 rows containing missing values or values outside the scale range
(`geom_interactive_point()`).
```

```
htmltools::save_html(interactive_plot, "/home/ryan/R/ggiraph-2.html")
```

10.11 Processing the data for comparison over time

In order to do something like study the change over time in the "Direction" question - or any of the other variables in this dataset - we have to do a little work to process the data beyond the raw counts we were using in the barcharts above.

In this example, the percentage of people thinking their country is moving in the right direction is computing by extracting the counts and discarding the NAs (only looking at the "Right" and "Wrong" answers). The code does this for each of the nine country and year combinations, and then pastes the results into a small matrix so we can plot the results.

This is one approach to solving the problem, among several other possibilities. For example, one could rename and restructure the data to make it easier to iterate over entries and automate some of this coding. One could also recode data to binary yes/no, 0-1 responses to make math on the proportions easier. In the approach below, the tidyverse package *forcats*, built to handle categorical data, makes an appearance. For my style of math brain, manually constructing the matrix seems the clearest way to handle this.

```
# compute percentages

my_final_data$Direction_recode <- as.factor(Direction_recode)

attach(my_final_data)
```

```
The following objects are masked from my_final_data (pos = 3):

    Age, China_Favor, Country, Country_Fin_Sit, Direction,
    Direction_recode, Education, Employment, Employment_recode,
    Employment_recode2, Family_Fin_Sit, Family_Fin_Sit_recode,
    Language, Main_News, Marital, Marital_recode, Remit, Russia_Favor,
    Sex, USA_Favor, Year


The following objects are masked from my_final_data (pos = 4):

    Age, China_Favor, Country, Country_Fin_Sit, Direction, Education,
    Employment, Family_Fin_Sit, Language, Main_News, Marital, Remit,
    Russia_Favor, Sex, USA_Favor, Year


The following objects are masked from my_final_data (pos = 5):

    Age, China_Favor, Country, Country_Fin_Sit, Direction, Education,
    Employment, Family_Fin_Sit, Language, Main_News, Marital, Remit,
    Russia_Favor, Sex, USA_Favor, Year
```

```r
mean(as.numeric(Direction_recode), na.rm=TRUE)
```

```
[1] 1.158222
```

```r
levels(my_final_data$Direction_recode)
```

```
[1] "Right direction" "Wrong direction"
```

```r
# forcats

fct_count(Direction_recode, sort = FALSE, prop = TRUE)
```

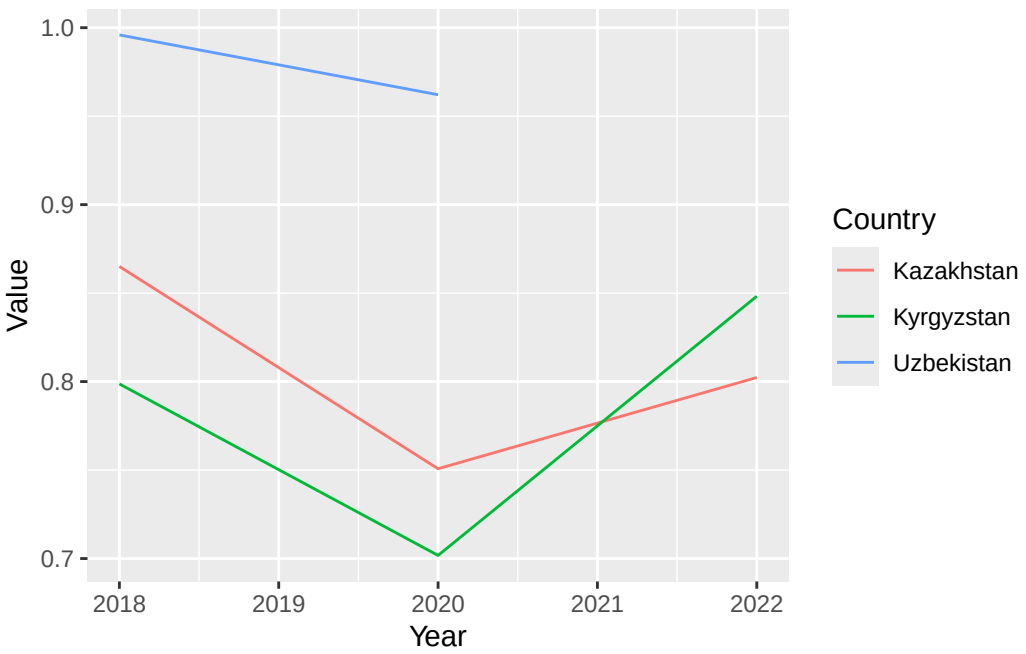
```
# A tibble: 3 x 3
  f                   n     p
  <fct>           <int> <dbl>
1 Right direction  9885 0.658
2 Wrong direction  1858 0.124
3 <NA>             3286 0.219
```

```r
mytable <- fct_count(Direction_recode, sort = FALSE, prop = FALSE)

Direction_percent <- mytable[1,2]/(mytable[1,2]+mytable[2,2])

# fct_match(Direction_recode, "Right direction")

# Now do a matrix of these

Uz18<- my_final_data |> filter(Country=="Uzbekistan" & Year==2018)
```

```r
Uz20<- my_final_data |> filter(Country=="Uzbekistan" & Year==2020)
Uz22<- my_final_data |> filter(Country=="Uzbekistan" & Year==2022)

Kz18<- my_final_data |> filter(Country=="Kazakhstan" & Year==2018)
Kz20<- my_final_data |> filter(Country=="Kazakhstan" & Year==2020)
Kz22<- my_final_data |> filter(Country=="Kazakhstan" & Year==2022)

Kg18<- my_final_data |> filter(Country=="Kyrgyzstan" & Year==2018)
Kg20<- my_final_data |> filter(Country=="Kyrgyzstan" & Year==2020)
Kg22<- my_final_data |> filter(Country=="Kyrgyzstan" & Year==2022)

# setup our matrix

Direction_pct_matrix <- data.frame(matrix(0,3,3))
rownames(Direction_pct_matrix) = c("Kazakhstan", "Kyrgyzstan", "Uzbekistan")
colnames(Direction_pct_matrix) = c("2018", "2020", "2022")

mytable <- fct_count(Kz18$Direction_recode, sort = FALSE, prop = FALSE)
Direction_pct_matrix[1,1] <- mytable[1,2]/(mytable[1,2]+mytable[2,2])

mytable <- fct_count(Kz20$Direction_recode, sort = FALSE, prop = FALSE)
Direction_pct_matrix[1,2] <- mytable[1,2]/(mytable[1,2]+mytable[2,2])

mytable <- fct_count(Kz22$Direction_recode, sort = FALSE, prop = FALSE)
Direction_pct_matrix[1,3] <- mytable[1,2]/(mytable[1,2]+mytable[2,2])

mytable <- fct_count(Kg18$Direction_recode, sort = FALSE, prop = FALSE)
Direction_pct_matrix[2,1] <- mytable[1,2]/(mytable[1,2]+mytable[2,2])

mytable <- fct_count(Kg20$Direction_recode, sort = FALSE, prop = FALSE)
Direction_pct_matrix[2,2] <- mytable[1,2]/(mytable[1,2]+mytable[2,2])

mytable <- fct_count(Kg22$Direction_recode, sort = FALSE, prop = FALSE)
Direction_pct_matrix[2,3] <- mytable[1,2]/(mytable[1,2]+mytable[2,2])

mytable <- fct_count(Uz18$Direction_recode, sort = FALSE, prop = FALSE)
Direction_pct_matrix[3,1] <- mytable[1,2]/(mytable[1,2]+mytable[2,2])

mytable <- fct_count(Uz20$Direction_recode, sort = FALSE, prop = FALSE)
Direction_pct_matrix[3,2] <- mytable[1,2]/(mytable[1,2]+mytable[2,2])

# this was the not asked question in Uzbekistan in 2022, so leave it alone,
 ↪  code as NA

mytable <- fct_count(Uz22$Direction_recode, sort = FALSE, prop = FALSE)

Direction_pct_matrix[3,3] <- NA
```

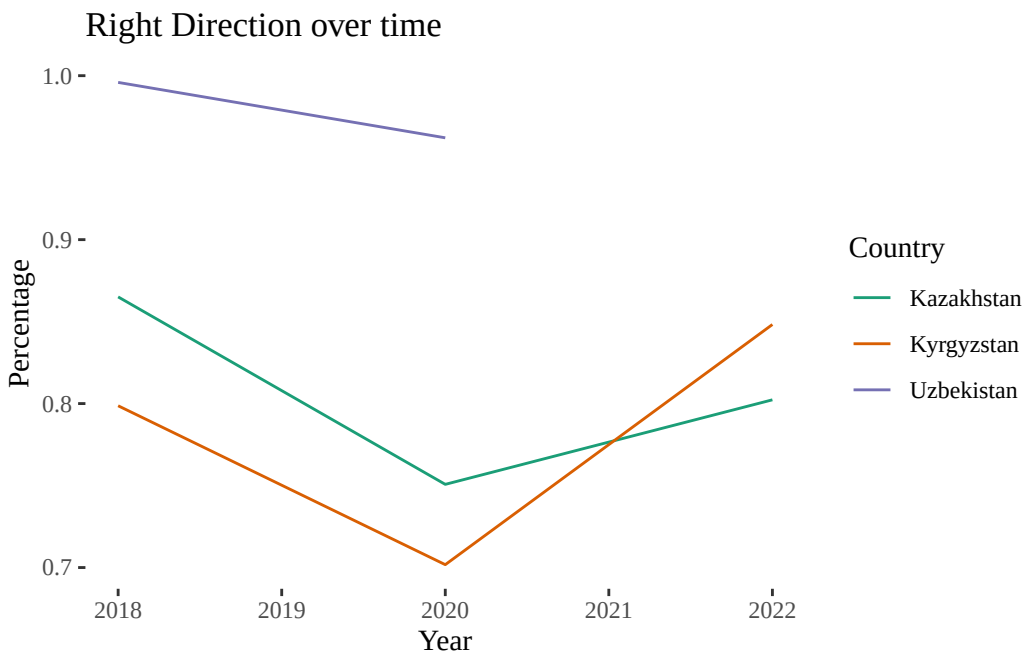
```
Direction_pct_matrix <- cbind(Direction_pct_matrix, Country=c("Kazakhstan",
 ↪  "Kyrgyzstan", "Uzbekistan"))

Direction_long <- pivot_longer(Direction_pct_matrix, !Country,
 ↪  names_to="Year", values_to = "Value")

Direction_long$Year <- as.numeric(Direction_long$Year)

ggplot(Direction_long, aes(x=Year, y=Value, color=Country)) +
  geom_line()
```

```
Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_line()`).
```




```
ggplot(Direction_long, aes(x=Year, y=Value, color=Country)) +
  geom_line() + theme_tufte() + scale_color_brewer(palette = "Dark2") +
  labs(title="Right Direction over time", y="Percentage")
```

```
Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_line()`).
```



11 Further statistical analysis

11.1 Stepwise regresssion

As another brief example of statistical analysis, we can try a stepwise regression to predict the answer to the right direction question. Logistic regression is the suitable method for when the outcome is a binary choice (yes/no, true/false, right/wrong).

```
lm(data=my_final_data, Remit ~ .)
```

```
Call:
lm(formula = Remit ~ ., data = my_final_data)

Coefficients:
                                                    (Intercept)
                                                      1.025e+02
                                                        SexMale
                                                     -5.143e+00
                                                            Age
                                                      8.998e-03
                                                  MaritalSingle
                                                     -3.198e+00
                                     MaritalWidowed or divorced
                                                     -3.078e+00
                                       EducationComplete school
                                                      2.140e+00
 EducationComplete vocational - vocational school, lyceum, etc.
```

-1.424e+00
EducationFinished school
3.207e+00
EducationIncomplete higher education
4.132e-01
EducationIncomplete school
1.371e+01
EducationIncomplete vocational education
3.607e+00
EducationNo education
2.024e+01
EducationVocational (vocational school, lyceum etc.)
2.742e+00
EmploymentA housewife / househusband
-7.413e+00
EmploymentA student
1.090e+00
EmploymentRetired
-1.561e+01
EmploymentUnemployed and actively seeking employment
1.931e+00
EmploymentUnemployed and not actively seeking employment
7.137e+00
EmploymentWorking full-time
-1.231e+01
EmploymentWorking part-time
-1.316e+01
DirectionWrong direction
5.883e-01
China_FavorRefused (vol.)
2.836e+01
China_FavorSomewhat Favorable
-4.770e-01
China_FavorSomewhat Unfavorable
7.221e+00
China_FavorVery Favorable
2.588e+00
China_FavorVery Unfavorable
7.166e-01
Russia_FavorRefused (vol.)
-8.448e+01
Russia_FavorSomewhat Favorable
-1.045e+00
Russia_FavorSomewhat Unfavorable
-4.756e+00
Russia_FavorVery Favorable
1.593e+00
Russia_FavorVery Unfavorable

-9.045e+00
USA_FavorRefused (vol.)
4.003e+01
USA_FavorSomewhat Favorable
-3.250e+00
USA_FavorSomewhat Unfavorable
-6.396e+00
USA_FavorVery Favorable
3.120e-01
USA_FavorVery Unfavorable
4.342e+00
Main_NewsFamily, friends, or neighbors
-3.346e+01
Main_NewsInternet
-3.057e+01
Main_NewsInternet: blogs, others
-3.226e+01
Main_NewsNon-government newspapers published in our country
-5.251e+01
Main_NewsNone / Not interested in news (vol.)
-4.714e+01
Main_NewsOther (vol.)
-1.381e+01
Main_NewsOur country's national television stations
-2.940e+01
Main_NewsOur government's newspapers
-4.035e+01
Main_NewsRadio
2.402e+01
Main_NewsReligious services or literature
-3.556e+01
Main_NewsRussian television stations
-5.486e+01
Main_NewsSocial media sites or messengers
-3.389e+01
LanguageKarakalplak
-1.140e+01
LanguageKazakh
-2.420e+01
LanguageKyrgyz
-2.518e+01
LanguageOther
-8.257e+01
LanguageOther (vol.)
-1.804e+01
LanguageRussian
-2.891e+01
LanguageTajik

```
                                                  6.440e+00
                                        LanguageTurkmen
                                                 -1.027e+02
                                          LanguageUzbek
                                                 -6.066e+00
                        Country_Fin_SitDon't Know (vol.)
                                                 -6.027e+00
                              Country_Fin_SitMuch Better
                                                 -1.649e+00
                               Country_Fin_SitMuch Worse
                                                  1.232e+01
                          Country_Fin_SitRefused (vol.)
                                                  3.010e+01
                         Country_Fin_SitSomewhat Better
                                                  4.447e+00
                          Country_Fin_SitSomewhat Worse
                                                  6.906e+00
                            Family_Fin_SitBecome Better
                                                 -6.082e+00
                             Family_Fin_SitBecome Worse
                                                 -7.769e+00
                              Family_Fin_SitMuch better
                                                 -5.003e+00
                               Family_Fin_SitMuch worse
                                                 -6.092e+00
                          Family_Fin_SitSomewhat better
                                                 -4.648e+00
                           Family_Fin_SitSomewhat worse
                                                 -4.229e+00
                          Family_Fin_SitStayed the Same
                                                 -5.070e+00
                                      CountryKyrgyzstan
                                                  1.216e+01
                                      CountryUzbekistan
                                                  8.194e+00
                                                   Year
                                                     NA
                                     Marital_recodeSingle
                                                     NA
                       Marital_recodeWidowed or divorced
                                                     NA
           Employment_recodeA housewife / househusband
                                                     NA
                              Employment_recodeA student
                                                     NA
                                Employment_recodeRetired
                                                     NA
Employment_recodeUnemployed and actively seeking employment
```

```
                                                                    NA
Employment_recodeUnemployed and not actively seeking employment
                                                                    NA
                              Employment_recodeWorking full-time
                                                                    NA
                              Employment_recodeWorking part-time
                                                                    NA
                                     Employment_recode2Housewife
                                                                    NA
                                       Employment_recode2Retired
                                                                    NA
                                    Employment_recode2Unemployed
                                                                    NA
                                       Employment_recode2Working
                                                                    NA
                                  Direction_recodeWrong direction
                                                                    NA
                                       Family_Fin_Sit_recodeSame
                                                                    NA
                                     Family_Fin_Sit_recodeWorse
                                                                    NA
```

```r
lm(data=my_final_data, Remit ~ Year)
```

```
Call:
lm(formula = Remit ~ Year, data = my_final_data)

Coefficients:
(Intercept)         Year
   4879.037       -2.389
```

```r
regression_output <- lm(Remit ~ Sex + Language, data=my_final_data)
summary(regression_output)
```

```
Call:
lm(formula = Remit ~ Sex + Language, data = my_final_data)

Residuals:
    Min      1Q  Median      3Q     Max
-61.499 -22.364  -1.316  20.770  70.193

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)             56.620     29.164   1.941 0.052443 .
SexMale                 -6.620      1.734  -3.817 0.000142 ***
LanguageKarakalpak      -6.238     30.218  -0.206 0.836483
```

```
LanguageKarakalplak       5.352     30.541   0.175 0.860916
LanguageKazakh          -14.301     29.763  -0.480 0.630972
LanguageKyrgyz           -5.303     29.160  -0.182 0.855724
LanguageOther           -30.000     41.171  -0.729 0.466347
LanguageOther (vol.)     -5.803     31.123  -0.186 0.852127
LanguageRefused (vol.)   -6.620     41.207  -0.161 0.872405
LanguageRussian         -20.193     29.293  -0.689 0.490751
LanguageTajik            -6.007     29.547  -0.203 0.838931
LanguageTurkmen         -41.080     33.636  -1.221 0.222212
LanguageUzbek             5.879     29.158   0.202 0.840237
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 29.11 on 1179 degrees of freedom
  (13837 observations deleted due to missingness)
Multiple R-squared:  0.08179,   Adjusted R-squared:  0.07244
F-statistic: 8.752 on 12 and 1179 DF,  p-value: 3.504e-16
```

```r
regression_output_all_demographic <- lm(Remit ~ Sex + Age + Language +
 ↪  Employment + Education + Country + Year, data=my_final_data)
summary(regression_output_all_demographic)
```

```
Call:
lm(formula = Remit ~ Sex + Age + Language + Employment + Education +
    Country + Year, data = my_final_data)

Residuals:
    Min      1Q  Median      3Q     Max
-62.284 -21.452  -0.478  20.883  74.774

Coefficients:
                                                                  Estimate
(Intercept)                                                      -71.97361
SexMale                                                           -3.68612
Age                                                                0.07580
LanguageKarakalpak                                               -17.81780
LanguageKarakalplak                                               -5.15660
LanguageKazakh                                                   -20.10909
LanguageKyrgyz                                                   -21.52807
LanguageOther                                                    -39.59865
LanguageOther (vol.)                                             -16.25928
LanguageRefused (vol.)                                           -17.19801
LanguageRussian                                                  -30.42818
LanguageTajik                                                    -15.26527
LanguageTurkmen                                                  -45.06972
LanguageUzbek                                                     -6.81348
EmploymentA housewife / househusband                             -12.57774
```

EmploymentA student	-6.50282
EmploymentDon't Know (vol.)	-25.14740
EmploymentRefused (vol.)	42.82873
EmploymentRetired	-20.65655
EmploymentUnemployed and actively seeking employment	-3.45304
EmploymentUnemployed and not actively seeking employment	-2.05662
EmploymentWorking full-time	-17.75971
EmploymentWorking part-time	-16.95468
EducationComplete school	-3.18996
EducationComplete vocational - vocational school, lyceum, etc.	0.38482
EducationFinished school	5.16304
EducationIncomplete higher education	0.87363
EducationIncomplete school	5.74694
EducationIncomplete vocational education	3.20953
EducationNo education	-11.19255
EducationVocational (vocational school, lyceum etc.)	2.93201
CountryKyrgyzstan	8.43419
CountryUzbekistan	2.44885
Year	0.07083
	Std. Error
(Intercept)	1704.63557
SexMale	1.91440
Age	0.07944
LanguageKarakalpak	29.71893
LanguageKarakalplak	30.14348
LanguageKazakh	28.76249
LanguageKyrgyz	28.65775
LanguageOther	40.47050
LanguageOther (vol.)	30.19320
LanguageRefused (vol.)	40.38517
LanguageRussian	28.47748
LanguageTajik	29.03213
LanguageTurkmen	32.92324
LanguageUzbek	28.65380
EmploymentA housewife / househusband	3.79413
EmploymentA student	4.39810
EmploymentDon't Know (vol.)	20.15057
EmploymentRefused (vol.)	28.29363
EmploymentRetired	3.91040
EmploymentUnemployed and actively seeking employment	4.05323
EmploymentUnemployed and not actively seeking employment	4.21557
EmploymentWorking full-time	3.41131
EmploymentWorking part-time	3.90417
EducationComplete school	3.06981
EducationComplete vocational - vocational school, lyceum, etc.	3.20755
EducationFinished school	3.59326
EducationIncomplete higher education	4.30015
EducationIncomplete school	5.43586

```
EducationIncomplete vocational education                          4.38735
EducationNo education                                            16.48783
EducationVocational (vocational school, lyceum etc.)              3.84373
CountryKyrgyzstan                                                 5.46578
CountryUzbekistan                                                 5.65653
Year                                                              0.84373
                                                                t value Pr(>|t|)
(Intercept)                                                      -0.042 0.966329
SexMale                                                          -1.925 0.054415
Age                                                               0.954 0.340186
LanguageKarakalpak                                               -0.600 0.548928
LanguageKarakalplak                                              -0.171 0.864200
LanguageKazakh                                                   -0.699 0.484603
LanguageKyrgyz                                                   -0.751 0.452677
LanguageOther                                                    -0.978 0.328053
LanguageOther (vol.)                                             -0.539 0.590330
LanguageRefused (vol.)                                           -0.426 0.670297
LanguageRussian                                                  -1.068 0.285518
LanguageTajik                                                    -0.526 0.599123
LanguageTurkmen                                                  -1.369 0.171286
LanguageUzbek                                                    -0.238 0.812089
EmploymentA housewife / househusband                             -3.315 0.000945
EmploymentA student                                              -1.479 0.139532
EmploymentDon't Know (vol.)                                      -1.248 0.212293
EmploymentRefused (vol.)                                          1.514 0.130369
EmploymentRetired                                                -5.282 1.52e-
07
EmploymentUnemployed and actively seeking employment             -0.852 0.394433
EmploymentUnemployed and not actively seeking employment         -0.488 0.625740
EmploymentWorking full-time                                      -5.206 2.28e-
07
EmploymentWorking part-time                                      -4.343 1.53e-
05
EducationComplete school                                         -1.039 0.298956
EducationComplete vocational - vocational school, lyceum, etc.    0.120 0.904525
EducationFinished school                                          1.437 0.151025
EducationIncomplete higher education                              0.203 0.839043
EducationIncomplete school                                        1.057 0.290628
EducationIncomplete vocational education                          0.732 0.464595
EducationNo education                                            -0.679 0.497377
EducationVocational (vocational school, lyceum etc.)              0.763 0.445736
CountryKyrgyzstan                                                 1.543 0.123082
CountryUzbekistan                                                 0.433 0.665150
Year                                                              0.084 0.933108

(Intercept)
SexMale                                                          .
Age
```

```
LanguageKarakalpak
LanguageKarakalplak
LanguageKazakh
LanguageKyrgyz
LanguageOther
LanguageOther (vol.)
LanguageRefused (vol.)
LanguageRussian
LanguageTajik
LanguageTurkmen
LanguageUzbek
EmploymentA housewife / househusband                          ***
EmploymentA student
EmploymentDon't Know (vol.)
EmploymentRefused (vol.)
EmploymentRetired                                             ***
EmploymentUnemployed and actively seeking employment
EmploymentUnemployed and not actively seeking employment
EmploymentWorking full-time                                   ***
EmploymentWorking part-time                                   ***
EducationComplete school
EducationComplete vocational - vocational school, lyceum, etc.
EducationFinished school
EducationIncomplete higher education
EducationIncomplete school
EducationIncomplete vocational education
EducationNo education
EducationVocational (vocational school, lyceum etc.)
CountryKyrgyzstan
CountryUzbekistan
Year
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 27.96 on 1158 degrees of freedom
  (13837 observations deleted due to missingness)
Multiple R-squared:  0.1682,    Adjusted R-squared:  0.1445
F-statistic: 7.096 on 33 and 1158 DF,  p-value: < 2.2e-16

backward_model <- step(regression_output_all_demographic, direction =
 ↪  "backward")

Start:  AIC=7973.94
Remit ~ Sex + Age + Language + Employment + Education + Country +
    Year

             Df Sum of Sq    RSS    AIC
- Education   8      4919 910094 7964.4
```

```
- Year        1         6 905181 7971.9
- Age         1       712 905887 7972.9
<none>                    905176 7973.9
- Country     2      3399 908575 7974.4
- Sex         1      2898 908074 7975.7
- Language   11     35971 941146 7998.4
- Employment  9     48953 954128 8018.7

Step:  AIC=7964.4
Remit ~ Sex + Age + Language + Employment + Country + Year

             Df Sum of Sq    RSS    AIC
- Age         1       691 910785 7963.3
<none>                    910094 7964.4
- Country     2      3330 913425 7964.7
- Year        1      2187 912281 7965.3
- Sex         1      2728 912822 7966.0
- Language   11     37583 947678 7990.6
- Employment  9     52830 962924 8013.7

Step:  AIC=7963.3
Remit ~ Sex + Language + Employment + Country + Year

             Df Sum of Sq    RSS    AIC
<none>                    910785 7963.3
- Country     2      3419 914204 7963.8
- Sex         1      2591 913376 7964.7
- Year        1      2619 913404 7964.7
- Language   11     37645 948430 7989.6
- Employment  9     57584 968369 8018.4
```

```r
# Print the summary of the selected model

summary(backward_model)
```

```
Call:
lm(formula = Remit ~ Sex + Language + Employment + Country +
    Year, data = my_final_data)

Residuals:
    Min      1Q  Median      3Q     Max
-63.585 -22.008  -0.736  21.355  75.324

Coefficients:
                                                          Estimate Std. Error
(Intercept)                                             2101.7939  1107.4362
SexMale                                                   -3.4720     1.9056
```

```
LanguageKarakalpak                                        -14.2290    29.6101
LanguageKarakalplak                                        -3.4237    29.9879
LanguageKazakh                                            -17.6569    28.6852
LanguageKyrgyz                                            -19.0046    28.5893
LanguageOther                                             -34.8426    40.0542
LanguageOther (vol.)                                      -12.5691    30.0969
LanguageRefused (vol.)                                    -11.1582    40.0508
LanguageRussian                                           -27.9790    28.4033
LanguageTajik                                             -12.9188    28.9693
LanguageTurkmen                                           -43.4653    32.8627
LanguageUzbek                                              -4.1061    28.5924
EmploymentA housewife / househusband                      -13.4788     3.7125
EmploymentA student                                        -7.6983     3.9334
EmploymentDon't Know (vol.)                               -23.7324    20.0477
EmploymentRefused (vol.)                                   44.4887    28.2298
EmploymentRetired                                         -19.3761     3.2252
EmploymentUnemployed and actively seeking employment       -4.1640     3.9998
EmploymentUnemployed and not actively seeking employment   -2.6289     4.1481
EmploymentWorking full-time                               -18.3147     3.3162
EmploymentWorking part-time                               -17.5594     3.8390
CountryKyrgyzstan                                           8.4415     5.4478
CountryUzbekistan                                           2.6419     5.6115
Year                                                       -1.0043     0.5483
                                                          t value Pr(>|t|)
(Intercept)                                                 1.898 0.057957 .
SexMale                                                    -1.822 0.068709 .
LanguageKarakalpak                                         -0.481 0.630931
LanguageKarakalplak                                        -0.114 0.909123
LanguageKazakh                                             -0.616 0.538318
LanguageKyrgyz                                             -0.665 0.506346
LanguageOther                                              -0.870 0.384541
LanguageOther (vol.)                                       -0.418 0.676302
LanguageRefused (vol.)                                     -0.279 0.780599
LanguageRussian                                            -0.985 0.324799
LanguageTajik                                              -0.446 0.655718
LanguageTurkmen                                            -1.323 0.186216
LanguageUzbek                                              -0.144 0.885835
EmploymentA housewife / househusband                       -3.631 0.000295 ***
EmploymentA student                                        -1.957 0.050564 .
EmploymentDon't Know (vol.)                                -1.184 0.236735
EmploymentRefused (vol.)                                    1.576 0.115308
EmploymentRetired                                          -6.008 2.51e-09 ***
EmploymentUnemployed and actively seeking employment       -1.041 0.298069
EmploymentUnemployed and not actively seeking employment   -0.634 0.526367
EmploymentWorking full-time                                -5.523 4.11e-08 ***
EmploymentWorking part-time                                -4.574 5.30e-06 ***
CountryKyrgyzstan                                           1.550 0.121529
CountryUzbekistan                                           0.471 0.637869
```

```
Year                                                    -1.832 0.067243 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 27.94 on 1167 degrees of freedom
  (13837 observations deleted due to missingness)
Multiple R-squared:  0.1631,    Adjusted R-squared:  0.1458
F-statistic: 9.473 on 24 and 1167 DF,  p-value: < 2.2e-16
```

12 Other possibilities

We haven't even touching on machine learning algorithms. Those will require even more careful data cleaning to be used effectively with this somewhat messy real-world data. But things like clustering, trees and random forests, and other methods might be able to effectively group the data, find patterns, and make predictions. See the tidymodels package mentioned below.

13 Conclusion

There is no one stopping point for data analysis. Keep going until you have a full understanding of the data and its interaction with the questions you seek to answer!

See Data Analysis 1 and Data Analysis 2 for further discussion of data analysis techniques.

13.1 R for Data Science

Consult the R for Data Science book to learn more about data wrangling. Also check out tidymodels a collection of packages for modeling and machine learning using tidyverse principles.